

FACULTAD DE INGENIERIA



UNMDP

Comprendiendo Inglés
Técnico-Científico

Inglés Técnico 1

EL SUSTANTIVO (Formación de plurales)

EJERCICIO 1.1

Escriba las formas plurales de los siguientes sustantivos:

battery	law
axis	phase
woman	branch
glass	nucleus
knife	medium
theory	dish
tooth	potato
number	box
window	score
criterion	leaf
bacillus	field

EL ARTÍCULO DEFINIDO "THE" Y EL INDEFINIDO "A" ("AN")

EJERCICIO 2.1

Traduzca las siguientes combinaciones

- | | |
|----------------------|-------------------------|
| 1) the energy | 8) the conditions |
| 2) the time | 9) the future |
| 3) the effects | 10) the engineer |
| 4) a result | 11) the classes |
| 5) an increase | 12) a switch |
| 6) a way | 13) the tv sets |
| 7) the engines | 14) a device |

EJERCICIO 2.2

Complete los espacios en blanco con "a" o "an" según corresponda y traduzca

- | | |
|-------------------------|------------------------|
| 1) property | 7) advance |
| 2) buzz | 8) example |
| 3) industry | 9) year |
| 4) axis | 10) window |
| 5) echo | 11) hour |
| 6) hero | 12) input |

GLOSARIO DEL EJERCICIO 1.1, 1.2, 1.3

<u>Palabra</u>	<u>Función</u>	<u>Significado</u>	<u>Otra función y significado</u>	
advance	s	avance	in advance = con anticipación/por adelantado	
axis	s	eje		
bacillus	s.	bacteria		
basis	s	fundamento, base		
battery	s	batería		
box	s	caja, cajón		
branch	s	rama, sucursal		
buzz	s	zumbido, chicharra		
charge	s	carga	v	cargar
cheap	adj	barato		
chief	s	jefe	adj	principal
child	s	niño, chico/a		
class	s	clase		
condition	s	condición		
criterion	s	criterio		
datum	s	dato		
device	s.	aparato, dispositivo		
dish	s	plato		
dynamo	s	dínamo, generador		
echo	s	eco		
energy	s	energía		
engine	s.	máquina, motor		
engineer	s	ingeniero/a		
example	s	ejemplo		
field	s.	campo, área		
flash	s	destello	v	destellar
foot	s	pie		
function	s	función	v	funcionar
future	s	futuro		
glass	s	vidrio (sust. incontable que puede usarse en plural), vaso (sust contable)		
half	s	mitad		
hero	s	héroe		
hour	s	hora		
increase	s	aumento, incremento		
industry	s	industria		
knife	s	cuchillo		
law	s	ley		
leaf	s	hoja		
life	s	vida		
man	s	hombre		
maximum	s	máximo		
medium	s	medio		
memory	s	memoria		
minimum	s	mínimo		
nucleus	s	núcleo		
number	s	número		
phase	s	fase		
phenomenon	s	fenómeno		
piano	s	piano		
potato	s	papa		
property	s	propiedad		
radio	s	radio		
radius	s	radio (de una circunferencia)		
result	s	resultado		
roof	s	techo		
sample	s	muestra		

score	s	puntaje, puntuación	v	tener puntaje
screen	s	pantalla		
sheet	s	hoja, lámina, chapa		
step	s	paso		step by step = paso a paso
switch	s	interruptor (llave)		
symbol	s	símbolo		
theory	s	teoría		
time	s	tiempo, momento		
tooth	s	diente		
TV set	s	televisor		
value	s	valor		
vocabulary	s	vocabulario		
way	s	modo, forma, manera, camino		
woman	s	mujer		
window	s	ventana		
year	s	año		

El ADJETIVO

EJERCICIO 3.1

Vuelva a escribir las frases dadas utilizando el antónimo correspondiente a cada adjetivo y traduzca

- 1) a long axis
.....
- 2) the good samples
.....
- 3) a fast change
.....
- 4) the new law
.....
- 5) a light leaf
.....
- 6) the tall men
.....
- 7) an easy theory
.....
- 8) the large screens
.....
- 9) a wide box
.....
- 10) the high score
.....
- 11) the important criteria
.....
- 12) a useful result
.....
- 13) the thick leaves
.....
- 14) the first effect
.....
- 15) a high concentration
.....

GLOSARIO DEL EJERCICIO 2.1, 2.2

<u>Palabra</u>	<u>Función</u>	<u>Significado</u>	<u>Otra función y significado</u>	
a	art	un, una		
advance	s	adelanto, anticipación	v	adelantar
buzz	s	zumbido, chicharra		
condition	s	condición		
echo	s	eco	v	producir eco
effect	s	efecto		
energy	s	energía		
engineer	s	ingeniero/a		
example	s	ejemplo		
hero	s	héroe		
input	s.	entrada		
future	s	futuro		
increase	s	aumento	v	aumentar
result	s	resultado	v	resultar
the	art	el, la, los, las		
time	s	tiempo, hora, vez, momento		
way	s	modo, manera, vía, camino		

GLOSARIO DEL EJERCICIO 3.1

<u>Palabra</u>	<u>Función</u>	<u>Significado</u>	<u>Otra función y significado</u>	
bad	adj	malo		
difficult	adj	difícil		
early	adj	temprano, primitivo, primeros		
easy	adj	fácil		
expensive	adj	caro		
fast	adj	rápido		
first	adj	primero		
good	adj	bueno		
heavy	adj	pesado		
high	adj	alto		
important	adj	importante		
large	adj	grande		
last	adj	último		
late	adj	tarde		
light	adj	liviano, luminoso	s	luz
long	adj	largo		
low	adj	bajo		
narrow	adj	angosto		
new	adj	nuevo		
old	adj	viejo		
short	adj	corto, bajo		
slow	adj	lento		
small	adj	pequeño		
tall	adj	alto		
thick	adj	grueso, espeso		
thin	adj	delgado		
unimportant	adj	no importante, sin importancia		
useful	adj	útil		

COMBINACIONES DE SUSTANTIVOS

EJERCICIO 4.1

Traduzca las siguientes combinaciones de sustantivos

- 1) energy *conversion*
- 2) pollution *control*
- 3) air *pollution*
- 4) water *power*
- 5) power *sources*
- 6) steam *engines*
- 7) key *difference*
- 8) growth *patterns*
- 9) population *growth*
- 10) food *consumption*
- 11) transportation *industry*
- 12) uranium *fuels*
- 13) fuel *supplies*
- 14) data *processing*
- 15) petroleum *resources*
- 16) irrigation *purposes*
- 17) gas *turbine*
- 18) jet *planes*
- 19) energy *input*
- 20) oil *wells*

EJERCICIO 4.2

Traduzca las siguientes combinaciones de sustantivos con dos o más sustantivos y/o adjetivos

- 1) pollution control *engineers*
- 2) energy conversion *devices*
- 3) electrical power *generation*
- 4) fossil-fuel-fired steam *engines*
- 5) radio and television *broadcasting*
- 6) world average per capita energy *consumption*
- 7) different energy use *sectors*
- 8) electrical generation and transportation *areas*
- 9) rotating-anode X-ray *tube*
- 10) English-speaking *country*
- 11) tool-and-die *maker*
- 12) long- and short-term *effects*
- 13) two-pole *switch*
- 14) 2-inch-diameter *pipes*
- 15) cloud- and web-based services
- 16) permanent-magnet moving-iron *instrument*
- 17) short-circuit feedback *admittance*
- 18) step-by-step *instructions*
- 19) signal-distortion test *set*
- 20) simple, low-current *purposes*

GLOSARIO DEL EJERCICIO 4.1

<u>Palabra</u>	<u>Función</u>	<u>Significado</u>	<u>Otra función y significado</u>	
air	s	aire		
consumption	s	consumo		
control	s	control		
conversion	s	conversión		
difference	s	diferencia		
engine	s	motor, máquina		
food	s	comida, alimento		
fuel	s	combustible		
gas	s	gas (a veces, gasolina)		
growth	s	crecimiento		
industry	s	industria		
input	s	entrada		
instrument	s	instrumento		
irrigation	s	irrigación		
jet	s	chorro		
oil	s	aceite, petróleo		
pattern	s	patrón, modelo		
petroleum	s	petróleo		
plane	s	plano, avión (airplane)		
pollution	s	polución		
population	s	población		
power	s	potencia, poder		
processing	s	procesamiento		
purpose	s	propósito, aplicación		
resource	s	recurso		
source	s	fuelle		
steam	s	vapor (de agua)		
supply	s	suministro, fuente		
transportation	s	transporte		
turbine	s	turbina		
uranium	s	uranio		
water	s	agua		
well	s	pozo	adv	bien

GLOSARIO DEL EJERCICIO 4.2

<u>Palabra</u>	<u>Función</u>	<u>Significado</u>	<u>Otra función y significado</u>	
admittance	s	admitancia		
and	conj	y		
anode	s	ánodo		
area	s	área		
average	s	promedio, medio		
based	adj.	basado		
broadcasting	s	transmisión		
capita	s	cápita, cabeza		
carrier	s	portador/a		
circuit	s	circuito		
cloud	s	nube		
country	s	país, campo		
current	s	corriente		
device	s	aparato, dispositivo, artefacto		
diameter	s	diámetro		
die	s	matriz	v	morir
distortion	s	distorsión		
electrical	adj	eléctrico		
English	s	inglés	adj	inglés
feedback	s	realimentación		

fire	v	alimentar, disparar	s	fuego
fossil	s	fósil	adj	fósil
generation	s	generación		
inch	s	pulgada		
instruction	s	instrucción		
iron	s	hierro		
magnet	s	imán		
maker	s	fabricante		
moving	adj	móvil		
pair	s	par		
per	prep	por		
permanent	adj	permanente		
pipe	s	caño		
pole	s	polo, poste		
ray	s	rayo		
rotating	adj	rotativo		
sector	s	sector		
set	s	juego, equipo, aparato, conjunto	v	colocar, fijar
signal	s	señal		
simple	adj	simple		
speaking	s	habla		
system	s	sistema		
television	s	televisión		
term	s	término, plazo		
test	s	ensayo, prueba	v	ensayar, probar
tool	s	herramienta		
tube	s	tubo		
two	num	dos		
use	s	uso		
web	s	red		
world	s	mundo		

EL VERBO “SER o ESTAR” (Presente)

EJERCICIO 5.1

Complete los espacios en blanco con “am”, “are” o “is” según corresponda y traduzca.

1) Canada and Australia _____ English-speaking countries.

.....

2) The young woman _____ tall.

.....

3) The first sample _____ good.

.....

4) John and William _____ electrical engineers.

.....

5) The 1-inch-diameter pipes _____ long.

.....

6) The world average per capita energy consumption _____ not high.

.....

7) An energy conversion device _____ useful.

.....

8) _____ gas turbines expensive?

.....

9) Tom and I _____ tool-and-die makers.

.....

10) The population growth _____ important.

.....

EJERCICIO 5.2

Reescriba las oraciones del Ejercicio 5.1 sustituyendo cada sujeto por el correspondiente pronombre y traduzca esta nueva versión.

1)

.....

2)

.....

3)

.....

4)

.....

- 5)
.....
- 6)
.....
- 7)
.....
- 8)
.....
- 9)
.....
- 10)
.....

EL VERBO “SER o ESTAR” (Pasado)

EJERCICIO 5.3

Complete los espacios en blanco con “was” o “were” según corresponda y traduzca las oraciones

- 1) It _____ a new permanent magnet.
.....
- 2) The heavy turbines _____ at the laboratory.
.....
- 3) The signal input _____ low.
.....
- 4) It _____ not a rotating device.
.....
- 5) _____ the oil industry important?
.....
- 6) It _____ cold yesterday.
.....
- 7) The test results _____ wrong.
.....
- 8) The new device _____ \$1m.
.....

GLOSARIO DEL EJERCICIO 5.1 y 6.1

am	v	soy, estoy
are	v	eres, sos, son, estás, están
cold	adj	frío
country	s	país
device	s	dispositivo, aparato
growth	s	crecimiento
input	s	entrada
is	v	es, está
magnet	s	imán
oil	s	petróleo
pipe	s	caño
speaking	adj	(de) habla
test	s	prueba, ensayo
was	v	pretérito del verbo to be
were	v	pretérito del verbo to be
wrong	adj	incorrecto
yesterday	adv	ayer

REPASO 1

Traduzca las siguientes oraciones

- 1) It is a good two-pole switch.
- 2) They are fast jet planes.
- 3) He is an important man.
- 4) They are not simple growth patterns.
- 5) Were they the first fuel supplies?
- 6) A steam engine is a useful device.
- 7) It was not a narrow screen.
- 8) They are short and thin women.
- 9) You are not an average engineer.
- 10) Is it an old energy resource?
- 11) Chemical engineering is an interesting field.
- 12) Was electrical power generation useful in the 1900s?
- 13) The pipes are 2 inches in diameter.
- 14) The weak positive to negative electron flow was short.
- 15) The good material properties are necessary.

REPASO 2

Traduzca las siguientes combinaciones de sustantivos

- | | |
|---------------------------------------|------------------------------------|
| 1) steam engine | engine steam |
| 2) power source | source power |
| 3) data processing control | control data processing |
| 4) irrigation-water supply | irrigation-supply water |
| 5) gas-turbine jet system | jet-turbine-gas system |
| 6) television signal input | television input signal |
| 7) transportation-fuel purposes | transportation-purpose fuels |
| 8) average consumption | consumption average |
| 9) car manufacturer | manufacturer car |
| 10) resource market | market resource |

REPASO 3

Traduzca las siguientes frases extraídas de una tabla de contenidos de un libro.

5

Air pollution concepts

- 5.1 INTRODUCTION
- 5.2 EFFECTS OF POLLUTION AND CLIMATE
- 5.3 EFFECTS OF POLLUTION ON PLANTS AND ANIMALS
- 5.4 SPECIFIC ATMOSPHERIC POLLUTION
- 5.5 SPECIFIC SCAVENGING¹ PROCESSES
- 5.6 AIR POLLUTION LEGISLATION

6

Engineering materials

- 6.1 MATERIALS IN ENGINEERING DESIGN
- 6.2 MECHANICAL TESTING
- 6.3 EXAMINATION OF MATERIALS
- 6.4 SELECTION OF MATERIALS
- 6.5 MATERIALS IN THE FUTURE
- 6.6 SUMMARY

7

Fundamentals of electrical networks

- 7.1 DEFINITIONS, CONCEPTS, AND CIRCUIT LAWS
- 7.2 THE BASIC CIRCUIT FORMS
- 7.3 TIME-VARYING SIGNALS
- 7.4 NONDISSIPATIVE PASSIVE ELEMENTS – INDUCTORS AND CAPACITORS
- 7.5 TRANSIENT² BEHAVIOR³ IN SERIES *RL* AND *RC* CIRCUITS
- 7.6 INSULATORS⁴
- 7.7 SUMMARY

8

Computer engineering

- 8.1 HISTORICAL PERSPECTIVE
- 8.2 GENERAL-PURPOSE DIGITAL COMPUTER
- 8.3 COMPUTER CODING AND NUMERICAL REPRESENTATIONS
- 8.4 SIGNALS IN LOGICAL SYSTEMS
- 8.5 DIGITAL LOGIC
- 8.6 THE MORGAN'S THEOREM
- 8.7 BINARY FUNCTIONS
- 8.8 COMPUTER SYSTEMS
- 8.9 SUMMARY

¹ limpiar, sanear

² transitorio, pasajero

³ comportamiento

⁴ aislante

EL TIEMPO PRESENTE SIMPLE

EJERCICIO 6.1

Complete los espacios con el Tiempo Presente Simple de los verbos dados entre paréntesis y traduzca.

1) Energy _____ living conditions more comfortable. (make)

.....

2) They _____ a different power source. (use)

.....

3) In chapter four we _____ the oil resources of the country. (discuss)

.....

4) The efficiency of the system _____ with a good design. (increase)

.....

5) It _____ an important television industry. (have)

.....

6) X-rays _____ the test time. (reduce)

.....

7) She often _____ at the control instruments. (look)

.....

8) I _____ several light tools. (need)

.....

9) The distortion _____ at a constant rate. (change)

.....

10) An electrical engineer _____ many useful properties. (know)

.....

11) Programmers _____ new applications. (develop)

.....

12) Useless catalysts _____ reaction time. (not reduce)

.....

13) A good environmental condition _____ test results. (not affect)

.....

14) Good signals _____ transmission quality. (not decrease)

.....

15) Engineers _____ buildings. (not design)

.....

EJERCICIO 6.2

Subraye la forma correcta de las dos opciones dadas entre paréntesis y traduzca las siguientes oraciones.

1) (Do/Does) engineers need steam engines?

.....

2) The efficiency of the turbine (do/does) not change.

.....

3) The rotating devices (do/does) not have a different power source.

.....

4) (Do/Does) the input signal increase at a constant rate?

.....

5) (Do/Does) good control instruments reduce the test time?

.....

6) Richard and I (do/does) not know the average energy consumption.

.....

7) Japan (do/does) not make thin 1-inch diameter pipes.

.....

8) (Do/Does) the feedback increase the distortion?

.....

9) Several students (do/does) not use the important data.

.....

10) (Do/Does) the country need new oil resources?

.....

GLOSARIO DEL EJERCICIO 6.1 y 6.2

<u>Palabra</u>	<u>Función</u>	<u>Significado</u>	<u>Otra función y significado</u>	
affect	v.	afectar		
at	prep	en, a		
building	s.	edificio		
catalyst	s.	catalizador		
change	v	cambiar, variar	s	cambio
chapter	s	capítulo		
comfortable	adj	confortable		
constant	adj	constante	s	constante
design	s	diseño	v	diseñar
develop	v.	desarrollar		
different	adj	diferente		
discuss	v	discutir, tratar		
efficiency	s	eficiencia, rendimiento		
environmental	adj.	ambiental		
have	v	tener (irreg. had, had)		
know	v	saber, conocer (irreg. knew, known)		
living	s	vida		
look	v	mirar	s	apariencia
make	v	hacer (irreg. made, made)		
many	adj	muchos		
more	adj	más	adv	más
need	v	necesitar	s	necesidad
of	prep	de		
often	adv	a menudo		
quality	s.	calidad		
rate	s	velocidad, ritmo, tarifa, índice	v	clasificar, tasar
reduce	v	reducir		
several	adv	varios		
signal	s.	señal		
test	s.	prueba, ensayo		
with	prep	con		

EL TIEMPO PASADO SIMPLE

EJERCICIO 7.1

Complete con el Tiempo Pretérito (indefinido o imperfecto) de los verbos dados entre paréntesis

Verbos regulares

- 1) We _____ an increase in the production of cars. (expect)
.....
- 2) The efficiency _____ close to 100% yesterday. (remain)
.....
- 3) They _____ the reserves into 3 main groups. (divide)
.....
- 4) It _____ the available supply of solar energy. (include)
.....
- 5) The last theory _____ the first law of thermodynamics. (consider)
.....
- 6) They _____ a different power source. (use)
.....
- 7) Useless catalysts _____ reaction time. (not reduce)
.....

Verbos irregulares

- 1) We _____ the different power source. (see)
.....
- 2) The first chapter _____ with "Signal Feedback". (begin)
.....
- 3) He _____ a slow water turbine. (choose)
.....
- 4) I _____ a new television set last week. (bring)
.....
- 5) The students _____ to the university by bus. (go)
.....
- 6) Energy _____ living conditions more comfortable. (make)
.....
- 7) The electrical engineer _____ many useful properties. (know)
.....

GLOSARIO DEL EJERCICIO 7.1

<u>Palabra</u>	<u>Función</u>	<u>Significado</u>	<u>Otra función y significado</u>	
available	adj	disponible		
begin	v	empezar (irreg. began, begun)		
bring	v	traer (irreg. brought, brought)		
bus	s	ómnibus		
by	prep	por	(by bus = en ómnibus)	
car	s	auto		
choose	v	elegir (irreg. chose, chosen)		
close	adj	cercano (close to =cercano a)	v	cerrar
consider	v	considerar		
divide	v	dividir		
expect	v	esperar		
go	v	ir (irreg. went, gone)		
group	s	grupo		
in	prep	en		
include	v	incluir		
into	prep	en, dentro de		
know	v.	conocer, saber		
main	adj	principal		
make	v.	hacer		
production	v	producción		
reduce	v.	reducir		
remain	v	permanecer		
reserve	s	reserva	v	reservar
see	v	ver (irreg. saw, seen)		
solar	adj	solar		
thermodynamics	s	termodinámica		
to	prep	a, hacia		
university	s	universidad		
use	v.	usar		
week	s	semana		
yesterday	adv	ayer		

EL TIEMPO FUTURO SIMPLE

EJERCICIO 8.1

Complete con el Tiempo Futuro Simple de los verbos dados entre paréntesis y traduzca.

1) We _____ the process tomorrow. (study)

.....

2) The fabrication _____ on the mechanical properties of the materials. (depend)

.....

3) You _____ vast environmental conditions. (find)

.....

4) It _____ an ideal situation. (be)

.....

5) I _____ the final gas pressure. (measure)

.....

6) He _____ several boxes in the laboratory. (leave)

.....

7) Charles and Jane _____ the first plane to New York. (take)

.....

8) They _____ all the new words on a paper. (write)

.....

9) We _____ to the selection of materials later. (return)

.....

10) The water temperature _____ between 15 and 20 degrees. (range)

.....

EJERCICIO 8.2

Traduzca las siguientes oraciones

1) The jet engine production will not decrease.

.....

2) Will they choose a university in advance?

.....

3) The sample will begin with a minimum value.

.....

4) We shall not consider the first results.

.....

5) Will men use the available fossil-fuel-fired steam engine?

.....

6) They will not discuss the long- and short-term effects.

.....

7) Will he change the permanent-magnet moving-iron instrument?

.....

8) Will they study property of the alloys?

.....

9) I shall not try the new sample.

.....

10) They won't get a good score in the exam.

.....

GLOSARIO DEL EJERCICIO 8.1

<u>Palabra</u>	<u>Función</u>	<u>Significado</u>	<u>Otra función y significado</u>	
all	adj	todo		
alloy	s.	aleación		
between	prep	entre		
degree	s	grado		
depend	v	depender (depend on/upon= depender de)		
environmental	adj	ambiental		
fabrication	s	fabricación		
final	adj	final		
find	v	encontrar (irreg. found, found)		
get	v.	obtener		
ideal	adj	ideal		
laboratory	s	laboratorio		
later	adv	más tarde, después		
leave	v	salir, dejar (irreg. left, left)		
material	s	material		
measure	v	medir		
mechanical	adj	mecánico		
on	prep	sobre, en		
paper	s	papel		
pressure	s	presión		
process	s	proceso		
range	v	oscilar, variar		
return	v	volver	s	retorno
score	s.	puntaje, puntuación		
selection	s	selección		
situation	s	situación		
study	v	estudiar	s	estudio
take	v	tomar, llevar (irreg. took, taken)		
temperature	s	temperatura		
tomorrow	adv	mañana	s	mañana
try	v.	probar		
vast	adj	vasto		
word	s	palabra		
write	v	escribir (irreg. wrote, written)		

REPASO 4

Traduzca las siguientes oraciones

1) The last chapter did not include the first law.

.....

2) It had a different growth pattern.

.....

3) Does the book begin with data processing?

.....

4) It did have an important property.

.....

5) Did you see the first carrier pattern?

.....

6) We knew the rate change of the admittance.

.....

7) They made large jet engines.

.....

8) They will have new resources.

.....

9) The energy supply did not remain constant.

.....

10) Did he expect a different buzz?

.....

11) It does not include simple, low current purposes.

.....

12) I did not look at the step-by-step instructions.

.....

13) Do you know a good tool-and-die maker?

.....

14) He does check the results in detail.

.....

15) I shall not work with the new engine.

.....

LOS VERBOS MODALES “CAN, MAY, MUST”

EJERCICIO 9.1

Traduzca las siguientes oraciones

1) He *must* design devices and structures.

.....

2) It *may* support the loads that you apply.

.....

3) Stainless steel *can* meet the design specifications.

.....

4) You *can* make a final choice of materials.

.....

5) The structure *must* fulfill two main requirements.

.....

6) He *may* come to the university tomorrow.

.....

7) They *must* be the first samples.

.....

8) The world population *may* have a fast increase in the future.

.....

9) She *can* design long pipe systems.

.....

10) We *may* study the process pattern.

.....

EJERCICIO 9.2

Traduzca las siguientes oraciones

1) Can they study the vast environmental conditions of the country?

.....

2) It may not be the ideal situation.

.....

3) Must it depend on the properties of the materials?

.....

4) He cannot take the first bus.

.....

5) May I write the new words on a sheet of paper?

.....

6) You must not return to the laboratory.

.....

7) Can she measure the final gas pressure?

.....

8) May the temperature range between 30 and 35 degrees tomorrow?

.....

9) I cannot find the main design specifications.

.....

10) The new students must not remain in the laboratory.

.....

GLOSARIO DEL EJERCICIO 9.1 y 9.2

<u>Palabra</u>	<u>Función</u>	<u>Significado</u>	<u>Otra función y significado</u>	
apply	v	aplicar		
can	v. modal	poder		
choice	s	elección		
country	s.	país		
environmental	adj	ambiental		
fulfill	v	cumplir, llenar		
load	s	carga	v	cargar
main	adj.	principal		
may	v. modal	puede que		
meet	v	reunir, encontrar, satisfacer (irreg. met, met)		
must	v. modal	deber		
range	v.	variar, oscilar		
requirement	s	requisito		
sheet	s.	papel		
steel	s	acero (stainless steel = acero inoxidable)		
support	v	soportar	s	soporte
take	v.	tomar		
vast	adj.	vasto		
that	pron rel	que	adj	ese/a, aquel/la

REPASO 5

Traduzca las siguientes oraciones

1) Is the car industry important?

.....

2) It's an old rotating device.

.....

3) The signal output must not be high.

.....

4) They're old steam turbines.

.....

5) It isn't a good permanent magnet.

.....

6) He knew many useless circuit theories.

.....

7) The buzz changed at a constant rate.

.....

8) You can't include the first law of thermodynamics.

.....

9) He divides the class into two groups.

.....

10) The students may not bring food.

.....

11) Did you choose a different carrier?

.....

12) They made several power sources last week.

.....

13) He often goes to the country.

.....

14) The living conditions were not good.

.....

15) It didn't need a new design.

.....

16) I was not at the control instruments.

.....

17) Does he expect a slow production growth?

.....

18) We found the ideal mechanical properties.

.....

19) A tall man measured the steam temperature.

.....

20) He must not remain close to the laboratory window.

.....

21) We shall not design large and heavy structures.

.....

22) She won't come to class tomorrow.

.....

23) They left New York at 5.00 yesterday.

.....

24) You cannot depend on the process selection.

.....

25) Did the X-ray tests begin?

.....

LA VOZ PASIVA

EJERCICIO 10.1

Lea el siguiente texto.

Energy Resources and Uses

The development of society can be characterized by a progressive substitution of machine power for muscle power.

The first attempt to use a power source occurred in the first century B.C. when water power was used for irrigation purposes. Even today, water is an important source of power, especially in mountainous terrain where electricity is generated in hydroelectric power stations.

During the Industrial Revolution the emphasis on energy resources was shifted from water power to fossil fuels. Water power was restricted to a few geographical areas but fossil-fuel-fired steam engines were mobile power sources.

The steam engine was first used as an auxiliary waterwheel pump and by the middle of the 19th century the steam engine became the principal power source for the manufacturing industry of the world.

The interest in energy consumption, energy reserves, and the ability to deliver energy where it is desired can be tied to industrialization. Thus, the great demands that are placed on the energy reserves of the earth can be explained by the fact that almost every country tries to industrialize – and industrialization takes energy.

For example, in the United States, the annual per capita energy consumption is approximately equal to the energy that can be obtained from 10 tons of coal.

EJERCICIO 10.2

Subraye las formas en VOZ PASIVA en el Ejercicio 10.1 y traduzca

EJERCICIO 10.3

Complete el siguiente cuadro de acuerdo con la secuencia de los eventos expuestos en el texto anterior (utilice una o 2 palabras en inglés).

<i>Century</i>	<i>Power Sources</i>	<i>Uses</i>
<i>Before 1st B.C.</i>		
<i>1st B.C.</i>		
<i>19th A.D.</i>		

EJERCICIO 10.4

En la tabla siguiente se enuncian algunas causas o consecuencias mencionadas en el texto del Ejercicio 11.1. Complete los cuadros en blanco en forma resumida de acuerdo al texto.

<i>Causes</i>	<i>Results</i>
<i>Development of a society</i>	
<i>Fossil-fuel-fired engines</i>	
<i>Water power was restricted to a few geographical areas</i>	
	<i>Interest in energy consumption, energy reserves, etc.</i>
	<i>Great demands are placed on energy reserves.</i>

EJERCICIO 10.5

Lea el siguiente texto.

Tension Test

In the tension test a specimen is subjected to increasing elongation until it fractures. The tensile load and elongation are measured at frequent intervals and the results are expressed in terms of stress and strain that are independent of the specimen size.

The normal stress is defined as the ratio of the load on the sample to the original cross-sectional area.

The average linear strain is defined as the ratio of the change in length of the sample to the original length.

Typical stress-strain curves are shown in figure 6.2

In the early stages of the tensile test the sample extends elastically, that is, the sample will return to the original length if the load is released. In the elastic region, stress is proportional to strain and is described by Hooke's law. The elastic modulus E defined by Hooke's law is very important in certain designs where the deformation must be kept to a minimum.

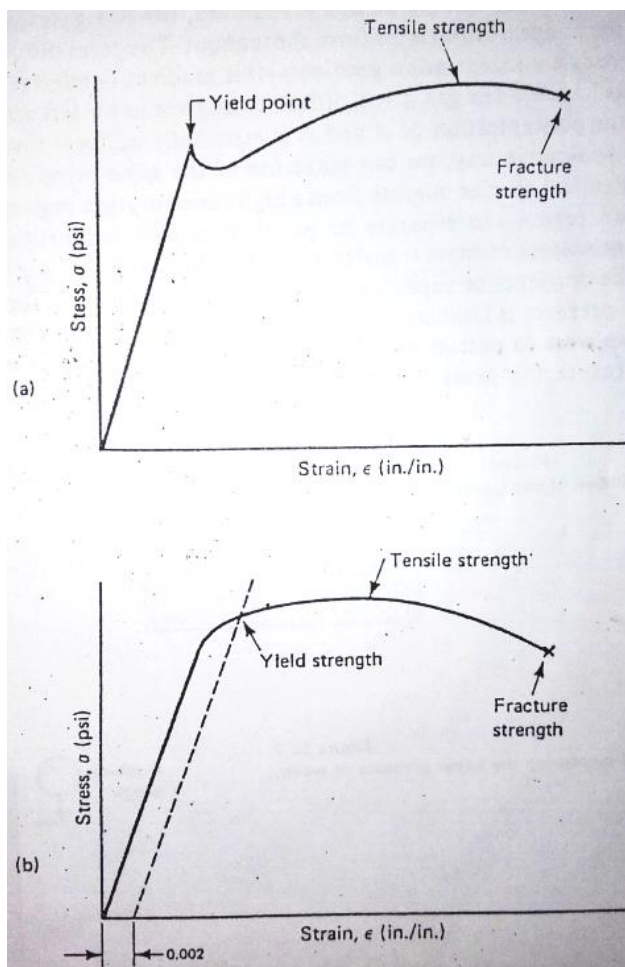


Figure 6.2

Typical tensile stress-strain curves.

(a) Low carbon steel. (b) Nonferrous metal.

EJERCICIO 10.6

Copie las formas en Voz Pasiva del 11.5 y escriba su equivalente en castellano con la forma "se..."

- | | | |
|-----------------|---|-----------|
| 1) is subjected | = | se somete |
| 2) | = | |
| 3) | = | |
| 4) | = | |
| 5) | = | |
| 6) | = | |
| 7) | = | |
| 8) | = | |
| 9) | = | |

GLOSARIO DEL EJERCICIO 10.1

<u>Palabra</u>	<u>Función</u>	<u>Significado</u>	<u>Otra función y significado</u>	
ability	s	habilidad, capacidad		
almost	adv	casi		
annual	adj	anual		
approximately	adv	aproximadamente		
as	conj	como	adv	a medida que
attempt	s	intento		
auxiliary	adj	auxiliar		
B.C	abrev	antes de Cristo (Before Christ)		
become	v	hacerse, volverse, convertirse (irreg. became, become)		
but	conj	pero, sino	prep	excepto
century	s	siglo		
characterize	v	caracterizar		
coal	s	carbon (mineral)		
deliver	v	entregar		
demand	s	demanda	v	demandar, exigir
desire	v	desear	s	deseo
development	s	desarrollo		
during	prep	durante		
earth	s	tierra		
electricity	s	electricidad		
emphasis	s	énfasis		
equal	adj	igual		
especially	adv	especialmente		
even	adv	incluso, aún	adj	parejo, par
every	adj	cada, todo		
example	s	ejemplo		
explain	v	explicar		
fact	s	hecho		
few	adj	pocos (a few = unos pocos)		
for	prep	para, por		
from	prep	de, desde		
generate	v	generar		
geographical	adj	geográfico		
great	adj	gran, grande		
hydroelectric	adj	hidroeléctrico		
industrial	adj	industrial		
industrialization	s	industrialización		
industrialize	v	industrializar		
interest	s	interés		
manufacturing	adj	manufacturero		
middle	s	mitad	adj	medio, central
mobile	adj	móvil		
mountainous	adj	montañoso		
muscle	s	músculo		
obtain	v	obtener		
occur	v	ocurrir, producir/se		
place	v	colocar	s	lugar
progressive	adj	progresivo		
pump	s	bomba	v	bombear
restrict	v	restringir		
revolution	s	revolución		
shift	v	cambiar, conmutar	s	cambio, turno
society	s	sociedad		
station	s	estación (power station = central de energía, estación eléctrica)		
substitution	s	sustitución		

terrain	s	terreno
thus	adv	así
tie	v	atar, ligar
today	adv	hoy
ton	s	tonelada
try	v	probar, intentar
waterwheel	s	rueda hidráulica
where	adv	donde, adonde

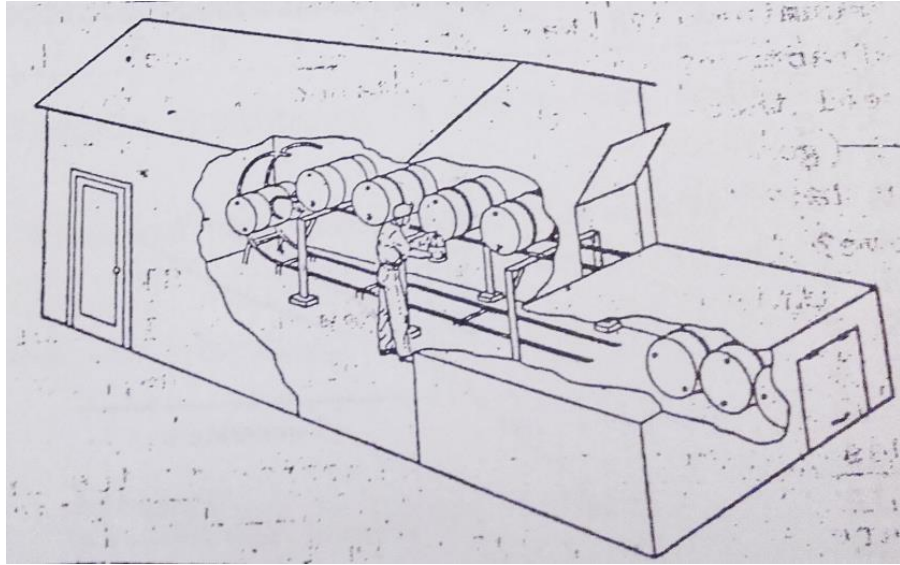
GLOSARIO DEL EJERCICIO 10.5

<u>Palabra</u>	<u>Función</u>	<u>Significado</u>	<u>Otra función y significado</u>	
cross-sectional	adj	transversal		
curve	s	curva		
define	v	definir		
deformation	s	deformación		
elastically	adv	elásticamente		
elongation	s	alargamiento		
express	v	expresar		
figure	s	figura, cifra	v	figurar, imaginar
fracture	v	fracturar		
frequent	adj	frecuente		
independent	adj	independiente		
interval	s	intervalo		
keep	v	mantener, conservar (irreg. kept, kept)		
length	s	longitud		
linear	adj	lineal		
modulus	s	módulo		
normal	adj	normal		
original	adj	original		
proportional	adj	proporcional		
ratio	s	cociente, relación		
region	s	región		
release	v	liberar		
show	v	mostrar (irreg. showed, shown)	s	espectáculo
size	s	tamaño, talla		
specimen	s	espécimen, probeta		
stage	s	etapa		
strain	s	deformación	v	estirar, tensar
stress	s	esfuerzo, tensión	v	someter a esfuerzo
subject	v	someter	s	sujeto, tema
tensile	s	tensión, tracción		
that is	expr	es decir		
typical	adj	típico		
until	conj	hasta	prep	hasta
very	adv	muy		
wire	s	cable, alambre		

PREPOSICIONES DE LUGAR

EJERCICIO 11.1

Subraye la preposición correcta de acuerdo con el dibujo y traduzca las oraciones.



1) The man is (*in, on, behind*) the house.

.....

2) 4 drums are placed (*in front of, above, below*) the man.

.....

3) The man is not (*in, on, at*) the door.

.....

4) The roof is (*over, below, by*) the man.

.....

5) I can see 3 drums (*over, on, above*) the floor and (*by, under, in*) a low door.

.....

6) The main door is (*behind, against, beside*) the man.

.....

7) He is standing (*by, on, over*) 3 drums that are (*over, above, on*) the upper rails.

.....

8) The picture does not show the sky or clouds (*on, above, over*) the house.

.....

9) The figure (*above, below, over*) was not colored.

.....

10) Drums roll (*of, from, under*) the upper (*to, at, in*) the lower rails.

.....

GLOSARIO DEL EJERCICIO 11.1

<u>Palabra</u>	<u>Función</u>	<u>Significado</u>	<u>Otra función y significado</u>	
above	prep	arriba, por encima de		
against	prep	contra		
behind	prep	detrás de		
below	prep	por debajo de		
beside	prep	al lado de, junto a		
cloud	s	nube		
color	v	colorear	s	color
door	s	puerta		
drum	s	tambor		
floor	s	piso		
house	s	casa		
in front of	prep	delante de		
lower	adj	inferior		
over	prep	sobre		
picture	s	figura, cuadro, película		
rail	s	riel		
roll	v	rodar		
sky	s	cielo		
under	prep	bajo, debajo de		
upper	adj	superior		

LOS TIEMPOS CONTINUOS O PROGRESIVOS

EJERCICIO 12.1

Complete los espacios con el Presente Progresivo de los verbos dados entre paréntesis.

- 1) They _____ the tensile load and the elongation. (measure)
.....
- 2) He _____ a very important law. (define)
.....
- 3) _____ you _____ the deformation to a minimum? (keep, interrog.)
.....
- 4) The sample _____ elastically during the last stage of the test. (extend, neg)
.....
- 5) The strain _____ independent of specimen size. (be)
.....
- 6) Some students _____ Ruby on Rails tutorial. (use)
.....
- 7) He _____ decisions with simplicity and flexibility. (make)
.....

EJERCICIO 12.2

Complete los espacios con el Pasado Progresivo de los verbos dados entre paréntesis.

- 1) They _____ an equal amount of energy from coal. (obtain)
.....
- 2) I _____ a new pump. (use, neg.)
.....
- 3) The development of society _____ more fuel reserves. (require)
.....
- 4) _____ you _____ the change in length) (consider, interrog.)
.....
- 5) She _____ physics at the university
.....
- 6) Some students _____ Ruby on Rails at university (develop)
.....
- 7) The company _____ the app to meet customer needs. (release)
.....

GLOSARIO DEL EJERCICIO 12.1, 12.2

<u>Palabra</u>	<u>Función</u>	<u>Significado</u>	Otra función y significado	
amount	s	cantidad	v	valer
physics	s	física		
pump	s.	bomba		
make decisions	exp.	tomar decisiones		
release	v.	lanzar		
require	v	requerir		
some	adj.	algunos/as		

CASOS DE -ING

<i>Cuando en inglés se usa una forma con -ing</i>	<i>Ejemplos</i>	<i>En castellano debe interpretarse:</i>
CASO 1: Con un TIEMPO PROGRESIVO	John is <i>loading</i> the printer. John was <i>loading</i> the printer	El correspondiente tiempo progresivo: John está <i>cargando la impresora</i> . John estaba <i>cargando la impresora</i> .
CASO 2: Después de una PREPOSICIÓN (excepto ON, IN, o de las conjunciones WHEN o WHILE)	We checked the meter <u>before</u> starting the test. They use steam <u>for</u> moving the turbine. We solved a problem <u>without</u> reading the previous theory.	El INFINITIVO Chequeamos el medidor <u>antes de</u> comenzar la prueba. Ellos usan vapor <u>para</u> mover la turbina. Solucionamos un problema <u>sin</u> leer la teoría previa
CASO 3: Después de las preposiciones ON/UPON, IN o de las conjunciones WHEN o WHILE.	<u>On arriving</u> we found that it was different. <u>In experimenting</u> with light bulbs Edison discovered the thermionic effect. <u>When experimenting</u> with light bulbs...	AL + INFINITIVO <u>Al</u> llegar encontramos que era diferente. <u>Al</u> experimentar con bombillas de luz Edison descubrió el efecto termiónico. <u>Al</u> experimentar con bombillas de luz...
CASO 4: Después de la preposición BY	We solved the problem <u>by using</u> logarithms. You can see the sign <u>by lifting</u> the cover. He will pass the exam <u>by studying</u> hard.	El PARTICIPIO PRESENTE (-ando/ -endo) pero omitiendo a preposición. Implica por medio de. Solucionamos el problema <i>usando</i> logaritmos. Podes ver el símbolo <i>levantando</i> la cubierta. El aprobará el examen <i>estudiando</i> duro.
CASO 5: Al actuar dicha forma como VERBO y SUSTANTIVO al mismo tiempo	<i>Smoking</i> is bad for you. One of my problems is <i>writing</i> letters. <i>Seeing</i> is believing	El INFINITIVO (o con el artículo "EL" + el INFINITIVO) (El) <i>fumar</i> es malo para vos. Uno de mis problemas es (el) <i>escribir</i> cartas. <i>Ver</i> es creer. (o Ver para creer)

CASO 6: Una PROPOSICIÓN ADJETIVA que ha sido reducida a una FRASE ADJETIVA. Se la reconoce porque la forma <i>-ing</i> le sigue inmediatamente al sustantivo que modifica. <u>Proposición</u>: conjunto de palabras relacionadas que tienen sujeto y verbo. <u>Frase</u>: conjunto de palabras relacionadas que no tienen sujeto y verbo	The cable <i>connecting</i> the amplifier is new (= The cable <u>that connects</u> the amplifier...) We proposed a plan <i>containing</i> 2 choices. (= We proposed a plan <u>that contained</u> ...) Materials <i>having</i> a high conductivity are very expensive (= Materials <u>that have</u> a high...)	El sustantivo + [QUE] + [VERBO CONJUGADO] (en el presente, pasado, etc.) según el tiempo implícito en la oración- El cable <i>que conecta</i> el amplificador es nuevo Propusimos un plan <i>que contenía</i> 2 opciones. Los materiales <i>que tienen</i> alta conductividad son caros.
CASO 7: A continuación de otro verbo. Por ej, FINISH, STOP, QUIT, DELAY, AVOID, INVOLVE, POSTPONE, KEEP, ENJOY, MIND, TRY, CONSIDER, DISCUSS, BEGIN, START, LIKE, PREFER, INTEND, OMIT, NEGLECT, CONTINUE, REMEMBER, requieren una forma <i>-ing</i> a continuación.	It <u>stopped</u> raining. We tried to <u>avoid</u> increasing the cost. It <u>began</u> raining. = It began to rain. He <u>likes</u> studying there. = He likes to study there.	El INFINITIVO (a veces precedido por la preposición <i>de, a, en, etc.</i>) <u>Paró</u> de llover. Intentamos <u>evitar</u> aumentar el costo. <u>Comenzó</u> a llover. Le <u>gusta</u> estudiar allí.
CASO 8: Como SUSTANTIVO	The <i>heating</i> of an alloy must be slow. We used arc <i>welding</i> to repair the pump. The instructions for correct paper <i>loading</i> are necessary.	El SUSTANTIVO El <i>calentamiento</i> de una aleación debe ser lento. Usamos <i>soldadura</i> de arco para reparar la bomba. Las instrucciones para la <i>carga</i> correcta del papel son necesarias.
CASO 9: Como ADJETIVO	The <i>resulting</i> vapors are not harmful. Some <i>escaping</i> electrons are attracted by the nearby atoms. They can measure the <i>moving</i> charges with a galvanometer	El ADJETIVO (-ante/-ente u otro), algunos casos deben considerarse como una combinación de sustantivo + sustantivo; o, como el caso 6. Los vapores <i>resultantes</i> no son dañinos. Algunos electrones <i>que se escapan</i> son atraídos por los átomos cercanos. Pueden medir las cargas <i>móviles</i> con un galvanómetro.

EJERCICIO 13.1

Complete los espacios con la forma -ing de los verbos dados entre paréntesis y traduzca. Indique la función que cumple en cada uno de los ejemplos

1) Laws _____ the behavior of electric charges will be developed. (govern)

.....

2) They must have a basic _____ of electric instruments. (understand)

.....

3) We shall consider the number of protons _____ to the right. (move)

.....

4) It may be understood by _____ the concepts defined above. (use)

.....

5) In _____ through the passive element, the current creates a voltage drop. (pass).

.....

6) A few jet aircrafts can make vertical takeoffs and _____. (land)

.....

7) A good utilization of _____ materials is very necessary. (exist)

.....

8) The _____ need of metals for high-temperature service will require new alloys.
(increase)

.....

9) The problem of _____ the right material for a given application is sometimes of
great complexity. (select)

.....

.....

10) In _____ the first law, the _____ convention is used for
_____ the algebraic sign of the voltage. (apply; follow; determine)

.....

.....

11) The _____ temperature is well above 1,000° C. (operate)

.....

12) It will be given before _____ the next section. (conclude)

.....

13) It is obtained by the passage of a fine X-ray beam through a powder of iron
_____ (file).

.....

14) You must consider the _____ interest in home computers. (grow)

.....

15) Numerical data _____ the thermal properties are obtained from the standard tests. (show)

.....

.....

EJERCICIO 13.2

Traduzca las oraciones y escriba a qué caso pertenecen las formas con ING subrayadas.

1) Steels show advantages as regards cost and recycling.

.....

2) In choosing a site, landscape considerations must be discussed.

.....

3) Floating constructions must be employed at sea.

.....

4) The topic of the conference is of increasing importance.

.....

5) Those plants generate revenues exceeding one billion dollars.

.....

6) While buying a suitable turbine, he chose a strong housing.

.....

7) He brought a few drawings showing the material creep.

.....

8) Hydropower is used almost exclusively for generating electricity in the country.

.....

GLOSARIO DEL EJERCICIO 13.1 y 13.2

<u>Palabra</u>	<u>Función</u>	<u>Significado</u>	<u>Otra función y significado</u>	
aircraft	s	aeronave		
algebraic	adj	algebraico		
alloy	s	aleación		
application	s	aplicación, solicitud		
basic	adj	básico		
beam	s	haz, rayo, viga		
before	prep	antes de, delante de		
behavior	s	comportamiento		
complexity	s	complejidad		
conclude	v	concluir		
convention	s	convención		
create	v	crear		
determine	v	determinar		
develop	v	desarrollar		
drop	s	caída, gota	v	dejar caer
exist	v	existir		
fine	adj	fino		
follow	v	seguir		
give	v	dar (irreg. gave, given)		
grow	v	crecer (irreg. grew, grown)		
home	s	hogar, casa		
land	v	aterrizar	s	tierra, territorio
metal	s	metal		
necessary	adj	necesario		
next	adj	próximo, siguiente		
numerical	adj	numérico		
operate	v	operar, funcionar, hacer funcionar		
pass	s	pase	v	pasar
passage	s	pasaje		
passive	adj	pasivo		
powder	s	polvo		
problem	s	problema		
proton	s	protón		
right	s	derecha, derecho	adj	derecho, correcto
select	v	seleccionar		
sign	s	signo, cartel	v	firmar
sometimes	adv	algunas veces		
standard	adj	normal, común	s	norma
takeoff	s	despegue (aviación)		
thermal	adj	térmico		
through	prep	a través de		
understand	v	entender		
utilization	s	utilización		
voltage	s	tensión (voltaje)		
advantage	s.	ventaja		
almost	adv.	casi		
as regards	frase adv.	con respecto a		
creep	s.	deformación por fluencia		
exceed	v.	exceder		
float	v.	flotar		
housing	s.	carcasa, cubierta		
hydropower	s.	energía hidráulica		
landscape	s.	paisaje		
recycle	v.	reciclar		
revenue	s.	ganancia		
suitable	adj.	apropiado/a/os/as		
topic	s.	tema		

ADJETIVOS: FORMAS CON –ING vs. –ED

Debe interpretarse bien la diferencia entre PARTICIPIOS PRESENTES y los PARTICIPIOS PASADOS usados como adjetivos. Los primeros califican a un sustantivo con una característica que está en proceso o que causa algo. Los participios pasados indican características ya adquiridas del sustantivo al que modifican.

PARTICIPIOS PASADOS (verbos regulares -ed)	PARTICIPIOS PRESENTES (-ing)
La terminación “-ed” como adjetivo corresponde en castellano a la terminación “-ado/-ido” del castellano, excepto en los casos irregulares en castellano.	La terminación “-ing” como adjetivo, se traducirá como: • primera opción: “-ante/-ente”. • segunda opción: un sustantivo en función de adjetivo unido al sust principal con una preposición, por ej., de, en, para. • tercera opción: caso 6, el tiempo verbal que se usa para traducirlo, dependerá del verbo principal de la oración.
<i>the escaped gas</i> = el gas <i>escapado</i> <i>a fascinated person</i> = una persona <i>fascinada</i> <i>an increased value</i> = un valor <i>aumentado</i> <i>the described procedure</i> = el procedimiento <i>descripto</i> <i>a written rule</i> = una regla <i>escrita</i> <i>the varied samples</i> = las muestras <i>variadas</i>	<i>the escaping gas</i> = el gas <i>que escapa/escapaba</i> <i>a fascinating person</i> = una persona <i>fascinante</i> <i>an increasing value</i> = un valor <i>que aumenta</i> <i>a growing pattern</i> = un patron <i>creciente /de crecimiento</i> <i>a writing device</i> = un dispositivo <i>de/para escritura</i> <i>the varying samples</i> = las muestras <i>variantes/que varían</i>

EJERCICIO 14.1

Traduzca las siguientes frases y compare

1) the expecting people

.....

2) the changing generation

.....

3) the extending country

.....

4) the moving particles

.....

5) this carrying device

.....

6) the determining function

.....

7) a controlling fact

.....

8) that landing plane

.....

9) the measuring interval

.....

10) those supporting structures

.....

the expected people

.....

the changed generation

.....

the extended country

.....

the moved particles

.....

this carried device

.....

the determined function

.....

a controlled fact

.....

that landed plane

.....

the measured interval

.....

those supported structures

.....

EL ADJETIVO DEMOSTRATIVO

EJERCICIO 15.1

Traduzca las siguientes oraciones.

1) It is beyond the scope of *this* section.

.....

2) Small oxide particles are dispersed through *these* alloys.

.....

3) Many people believe that *this* will be the material of the future.

.....

4) *Those* concepts are explained in *this* book.

.....

5) Does he understand *that* theory?

.....

6) It was left on that surface during the process

.....

7) Those mechanical properties will be found.

.....

8) These theories were not known before.

.....

9) More fuel resources are required by those developed countries.

.....

10) These developing countries are having a growing need of power stations.

.....

EJERCICIO 15.2

Subraye la forma correcta y traduzca cada oración

1) (*That/Those*) axis was drawn at 90 degrees.

.....

2) Did (*that/those*) men bring (*this/these*) tools?

.....

3) (*This/These*) is not the color that I chose.

.....

4) (*That/Those*) microwave devices were made in Japan.

.....

5) Things were not always (*this/these*) way.

6) (*This/these*) problem is being understood by many students now.

7) (*That/those*) book was written in the last century.

EJERCICIO 15.3

Traduzca las siguientes oraciones

1) He selected these alloys but then he used *that one*.

2) You must put a + sign before this relationship and a – sign before *that one*.

3) Can't they see *that*?

4) *This* must be understood first.

5) *Those* were perfect landings but I prefer *this one*.

6) A fast carrier was chosen at the beginning, but then they chose *that one*.

7) Was *that one* shown in the first figure?

GLOSARIO DE LOS EJERCICIOS 15.1, 15.2 y 15.3

<u>Palabra</u>	<u>Función</u>	<u>Significado</u>	<u>Otra función y significado</u>	
beyond	prep	más allá		
disperse	v	dispersar		
microwave	s	microonda		
oxide	s	óxido		
particle	s	particular		
people	s	gente, personas, pueblo		
perfect	adj	perfecto		
prefer	v	preferir		
relationship	s	relación		
scope	s	alcance		
that	adj dem	ese, esa, aquel, aquella	pron	eso, aquello
these	adj dem	estos, estas	pron	éstas, estos
this	adj dem	este, esta	pron	esto
those	adj dem	esos, esas, aquellos, aquellas	pron	aquellos, aquellas

REPASO 6 (actividad opcional)

A- Traducir las siguientes oraciones e identificar cada caso de -ing.

1. Educating the consumer is the focus of this book presenting the challenges in switching to new sources.

.....
.....

2. The book is not about global warming but it shows various ways of changing our energy generation for emitting very little CO₂.

.....
.....

3. New designs and electric engines can reduce transportation to 20Kwh and heating and cooling can be cut to 30kwh per person per day.

.....
.....

4. The improvements in areas like lighting will be offset by new uses of electricity and some arguments that usage will not decrease are very convincing.

.....
.....

5. The author does a great job when explaining the challenges that come with sources that do not provide energy on demand 24 hours a day.

.....
.....

6. We are getting a modest amount of our power from these sources; however, there is no magic storing technology today that allows to store extra power.

.....
.....

7. He did not mention geoengineering techniques to delay or reduce the effects of global warming, nor about other gases that cause global warming.

.....
.....

8. The amount of government R&D going into energy is surprisingly low; the UK is spending less than £0.2 per person per year.

.....
.....

9. People need to understand what is going on and be part of the process of moving the world to a new energy infrastructure.

.....
.....

GLOSARIO DEL REPASO 6

<u>Palabra</u>	<u>Función</u>	<u>Significado</u>	Otra función y significado
about	prep.	sobre, acerca de	
allow	v.	permitir	
amount	s.	cantidad	
argument	s.	argumento	
challenge	s.	desafío	
convince	v.	convencer	
cool	v.	refrigerar	
cut	v	recortar, reducir	
decrease	v.	disminuir, decrecer	
delay	v.	retrasar	
emit	v.	emitir	
focus	s.	foco	
get	v.	obtener	
get implemented	v	implementarse	
go into	v.	destinar	
go on	v.	suceder	
great	adj	gran	
heat	v.	calentar	
however	adv	sin embargo	
improvement	s.	mejora	
job	s.	trabajo	
less	adj.	menos	
light	v.	iluminar	
little	adj.	poco	
nor	conj.	ni	
move	v.	cambiar	
offset	v.	compensar	
over	adv.	más de	
provide	v	proveer	
R&D	s	Research and Development – investigación y Desarrollo (I + D)	
reduce	v.	reducir	
regulation	s.	regulación, ley	
some	adj.	algunos/as	
source	s	fuelle	
spend	v.	gastar	
store	v.	almacenar	
switch	v.	cambiar	
take	v	llevar	
understand	v.	entender	
usage	s.	uso	
various	adj.	distinto, diferente	
very	adv.	muy	
warm	s.	calentar	
way	s.	modo, forma, manera, camino	
what	pron	qué	

EXPRESIONES DE PROPÓSITO

EJERCICIO 16.1

Por cada oración dada escriba otras dos formas que expresen propósito y traduzca.

1) It is done to change the value of the variable over the course of an equation

a.

b.

c.

Traducción:

2) That lever is pulled toward you for loading the paper

a.

b.

c.

Traducción:

3) He stopped the printer in order to slide the guide to the left

a.

b.

c.

Traducción:

4) You must push the lever away from you to set the paper bail.

a.

b.

c.

Traducción:

5) To shift the loaded sheet to the right you must use that knob.

a.

b.

c.

Traducción:

EJERCICIO 16.2

Traduzca las siguientes oraciones

1) It gives a sample program **to** incorporate many of the techniques and commands discussed in this section.

.....

.....

2) You can edit the program **for** changing the assignments in those lines.

.....

.....

3) Parentheses are used **to** tell the computer to add the numbers before it divides.

.....

.....

4) Line 40 adds the variables and divides by 3 **in order to** get the average.

.....

.....

5) **To** print numbers, use the PRINT command followed by the number(s) you want to print.

.....

.....

6) You must type the following statement **for** returning the computer to the starting state.

.....

.....

7) **In order to** do this, you may press a function key or the space bar.

.....

.....

8) The program below is run **to** see how the ON command works.

.....

.....

9) This task is **for** accessing and maintaining files on disks.

.....

.....

10) This technology follows some development principles **to** avoid repeating code.

.....

.....

GLOSARIO DEL EJERCICIO 16.1

<u>Palabra</u>	<u>Función</u>	<u>Significado</u>	<u>Otra función y significado</u>	
away	adv	afuera, dirección contraria		
bail	s	barra de sujeción		
course	s	curso		
in order to	expr	para (propósito)		
knob	s	manija, perilla, picaporte		
left	s	izquierda	adj	izquierdo
lever	s	palanca		
pull	v	tirar	s	tiro
push	v	empujar	s	empuje
slide	v	deslizar/se (irreg. slid, slid)	s	deslizamiento
stop	v	parar, detener	s	parade
toward	prep	hacia		
variable	s	variable	adj	variable

GLOSARIO DEL EJERCICIO 16.2

<u>Palabra</u>	<u>Función</u>	<u>Significado</u>	<u>Otra función y significado</u>	
access	v	acceder		
add	v	agregar, sumar		
appropriate	adj	apropiado		
assignment	s	tarea		
bar	s	barra, bar		
command	s	comando, orden	v	ordenar
count	v	contar		
disk	s	disco		
edit	v	editar		
file	s	archivo, lima	v	archivar, limar
get	v	obtener, conseguir, etc. (irreg. got, got o gotten)		
graphic	s	gráfico		
how	adv	como		
idea	s	idea		
incorporate	v	incorporar		
line	s	línea		
maintain	v	mantener		
make sure	expr	asegurarse		
mode	s	modo		
or	conj	o		
parentheses	s	paréntesis		
press	v	presionar, oprimir, apretar		
program	s	programa		
run	v	ejecutar, hacer funcionar (irreg. ran, run)		
space	s	espacio	v	separar
statement	s	enunciado, oración		
task	s	tarea		
technique	s	técnica		
tell	v	decir (irreg. told, told)		
type	v	escribir (a máquina), tipear	s	tipo
work	v	trabajar, funcionar	s	trabajo

OTROS USOS DE “to + infinitivo”

EJERCICIO 16.3

Traduzca las siguientes oraciones

1) The next step is **to** select the appropriate graphic mode.

.....

2) The PRINT command tells the computer **to** display information on the screen.

.....

.....

3) To print numbers, use the PRINT command followed by the number(s) you want **to** print.

.....

.....

4) This book will teach the user how **to** program in the BASIC Programming Language.

.....

.....

5) Scientists began **to** investigate how and why substances react.

.....

.....

6) Its ultimate goal is to deliver the value quickly.

.....

7) Thus, the great demands that are placed on the energy reserves of the earth can be explained by the fact that almost every country tries **to** industrialize.

.....

.....

8) The concatenation, especially of long strings, first requires **to** allocate enough memory.

.....

.....

9) It's a good idea **to** count the number of left parentheses and right parentheses to make sure they are equal.

.....

.....

10) It is advisable **to** use a cloud environment in all operating systems.

.....

.....

GLOSARIO DEL EJERCICIO 16.3

<u>Palabra</u>	<u>Función</u>	<u>Significado</u>	Otra función y significado
advisable	adj.	aconsejable	
allocate	v	asignar	
as... adj ... as	exp.	tan... (adj) como	
began	pret.	begin (comenzar)	
chose	pret.	choose (elegir)	
count	v	contar	
dangerous	adj	peligroso	
deliver	v	suministrar	
display	v	mostrar	
enough	adj	suficiente	
every	adj	cada	
fact	s	hecho	
file	s	archivo	
goal	s	objetivo	
hope	v	esperar	
how	pron interrog	cómo	
its	adj pos	su/s	
left	s	izquierda	
make sure	v	asegurarse	
next	adj	próximo, siguiente	
pass	v	aprobar	
print	v	imprimir	
right	s	derecha	
screen	s	pantalla	
step	s	paso	
string	s	cadena	
teach	v	enseñar	
ultimate	adj	final	
want	v	querer	
why	pron interrog	por qué	

EL IMPERATIVO

EJERCICIO 17.1

Lea el siguiente texto

Rules for Typing BASIC Language Programs

You can type and use BASIC language programs even without knowing BASIC. You must type carefully, however, because a typing error may cause the computer to reject your information. The following guidelines will help minimize errors when typing or copying a program listing.

- 1- Spacing between words is not critical; e.g., typing FORT=1TO10 is the same as typing FOR T = 1 TO 10. However, a BASIC keyword itself must not be broken up by spaces. (See the BASIC 7.0 encyclopedia in Chapter V for a list of BASIC keywords).
- 2- Any characters can be typed inside quotation marks. Some characters have special functions when placed inside quotation marks. These functions are explained later in this Guide.
- 3- Be careful with punctuation marks. Commas, colons and semicolons also have special properties, explained later in this section.
- 4- Always press the RETURN key (indicated in this Guide by **RETURN**) after completing a numbered line.
- 5- Never type more than 160 characters in a program line. Remember, this is the same as four full screen lines in 40-column format, or two full screen lines in 80-column format. See section 8 for more details on 40- and 80-column formats.
- 6- Distinguish clearly between the letter I and the numeral 1 and between the letter O and the numeral 0.
- 7- The computer ignores anything following the letters REM on a program line. REM stands for REMark. You can use the REM statement to put comments in your program that tell anyone listing the program what is happening at a specific point.

Follow these guidelines when you type the examples and programs shown in this section.

The PRINT command tells the computer to display information on the screen. You can print both numbers and text (letters), but there are special rules for each case, described in the following paragraphs.

Printing Numbers

To print numbers, use the PRINT command followed by the number(s) you want to print. Try typing this on your Commodore 128.

PRINT 5

Then press the RETURN key. Notice the number 5 is now displayed on the screen.

Printing Text

Now that you know how to print numbers, it's time to learn how to print text. It's actually very simple. Any words or characters you want to display are typed on the screen, with a quote symbol at each end of the string of characters. **String is the BASIC name for any set of characters surrounded by quotes.** The quote character is obtained by pressing SHIFT and the numeral 2 on the top row of the keyboard (not the 2 in the numeric keypad). Try these examples:

PRINT "COMMODORE 128" RETURN

PRINT "4*5" RETURN

EJERCICIO 17.2

Lea cuidadosamente las instrucciones dadas a continuación para escribir un programa en Lenguaje BASIC. Decida si, de acuerdo al texto anterior, cada línea de instrucciones es Correcta, Incorrecta, o No Incluida en las consideraciones del texto. Coloque "C", "I" o "NI" delante del número de cada instrucción según su elección.

- 1) Type the first program line in this way:

```
10 PRINT "THIS IS A SAMPLE PROGRAM"
```

- 2) Do not press the RETURN key after completing line 10 and begin the next program line as follows:

```
20 FOR X=1 TO 5
```

- 3) Press the RETURN key and write:

```
30 PRINT "LET'S SEE IF THE 203-CHARACTER STATEMENT IN THIS LINE  
CAN BE PRINTED 5 TIMES ACCORDING TO THE COMMAND GIVEN IN  
LINE NUMBER 20. IF NOT, TRY AGAIN PLACING A COLON (:) OR A  
SEMICOLON (;) AFTER THE LAST WORD"
```

- 4) Now type:

```
40 NEXT X: REM "DON'T FORGET THIS LINE IF YOU WANT TO HAVE LINE  
30 PRINTED 5 TIMES"
```

- 5) In order to remember who wrote this program add the following:

```
50 REM "WRITTEN BY JOHN SMITH"
```

Let's not worry. The words written in the above line will not appear when you run the program.

EJERCICIO 17.3

Subraye todas las formas imperativas de los Ejercicios 17.1 y 17.2

GLOSARIO DEL EJERCICIO 17.1

<u>Palabra</u>	<u>Función</u>	<u>Significado</u>	<u>Otra función y significado</u>	
actual	adj	actual, verdadero		
any	adj	cualquier (afir), algún (interr), ningún (neg)		
anyone	adj	cualquiera (afir), alguien (interr), nadie (neg)		
anything	adj	cualquier cosa (afir), algo (interr), nada (neg)		
because	conj	porque		
both	adj	ambos	pron	ambos, tanto... como...
break	v	romper, quebrar (irreg. broke, broken)		break up = separar
	s	pausa, descanso		
careful	adj	cuidadoso		
carefully	adv	cuidadosamente		
case	s	caso		
cause	v	causar, hacer (que)		
character	s	caracter (letra), carácter		
clear	adj	claro		
colon	s	dos puntos (:)		
column	s	columna		
comma	s	coma (,)		
comment	s	comentario	v	comentar
complete	v	completar	adj	completo
copy	s	copia	v	copiar
critical	adj	crítico		
describe	v	describir		
display	v	mostrar, exhibir	s	muestra, exhibición
distinguish	v	distinguir		
e.g.	abrev	por ejemplo (abrev. de exempli gratia)		
each	adj	cada	pron	cada uno
encyclopedia	s	enciclopedia		
end	s	extremo, fin	v	finalizar
error	s	error		
format	s	formato		
four	num	cuatro		
full	adj	lleno, completo		
guide	s	guía	v	guiar
guideline	s	pauta, directiva		
happen	v	suceder		
help	v	ayudar	s	ayuda
however	adv	sin embargo		
ignore	v	ignorar		
indicate	v	indicar		
information	s	información		
inside	adv	dentro de, adentro	adj	interior
itself	adj réflex	en sí, propiamente dicho		
keypad	s	bloque de teclas, teclado		
keyword	s	palabra clave		
language	s	idioma, lenguaje		
learn	v	aprender		
letter	s	letra, carta		
list	v	listar	s	lista
mark	s	marca, símbolo	v	marcar
minimize	v	reducir, minimizar		
notice	v	observar	s	atención, anuncio
numeral	s	numeral		
numeric	adj	numérico		
paragraph	s	párrafo		
point	s	punto		
print	v	imprimir		

punctuation	s	puntuación		
quotation marks		comillas		
quote	s	comilla	v	citar
reject	v	rechazar		
remark	s	acotación, comentario		
row	s	fila		
rule	s	regla		
same	adj	mismo/a (the same as = lo mismo que)		
semicolon	s	punto y coma (;)		
special	adj	especial		
specific	adj	específico		
stand for	v	representar, significar		
string	s	cadena (en computación), cuerda		
surround	v	rodear, circundar		
text	s	texto		
than	conj	que (en comparativo)		
then	adv	luego, entonces	conj	entonces, en ese caso
top	s	parte superior		
want	v	querer		
what	pron rel	lo que,	pron interr.	qué, cual
when	adv	cuando	pron interr.	Cuando
without	prep	sin		
your	adj pos	su (de Ud., de Uds.)		

VERBO "THERE BE"

EJERCICIO 18.1

Complete los espacios en blanco con "there is" o "there are" y traduzca.

1) _____, of course, an exception to the previous statement.

.....

2) _____ many engineers interested in heat engine design, oil refining, and manufacturing.

.....

.....

3) _____ much sulfur dioxide in the stack gas? (interrog.)

.....

4) _____ a conversion of carbon to either carbon dioxide or carbon monoxide.

.....

5) _____ few power plant furnaces running at very low or zero excess air.

.....

EJERCICIO 18.2

Traduzca las siguientes oraciones

1) *There will be* a tendency for smoke formation during the process.

.....

2) *There may not be* a change in the value of the variable.

.....

3) *There were* small percentages of CO₂, SO₂, and ash in the combustion.

.....

4) *There must not be* a program line with 180 characters or more

.....

5) According to the law of conservation of matter *there cannot be* mass created or destroyed.

.....

GLOSARIO DE LOS EJERCICIOS 18.1 y 18.2

<u>Palabra</u>	<u>Función</u>	<u>Significado</u>	Otra función y significado
according to	adv	de acuerdo con/a, según	
ash	s	ceniza	
carbon	s	carbón	
combustion	s	combustión	
conservation	s	conservación	
destroy	v	destruir	
dioxide	s	dióxido, anhídrido	
either... or...	expr	either no se traduce, indica que se van a enumerar opciones) ... o	
exception	s	excepción	
excess	s	exceso	
formation	s	formación	
furnace	s	horno (industrial), caldera	
heat	s	calor	
manufacture	v	fabricar	
mass	s	masa	
matter	s	materia	
monoxide	s	monóxido	
much	adj	mucho	
of course	expr	por supuesto	
percentage	s	porcentaje	
previous	adj	previo	
refine	v	refinar	
smoke	s	humo	
stack	s	chimenea	
sulfur	s	azufre	
tendency	s	tendencia	
there are	v	hay (presente del verbo "there be" usado delante de formas plurales)	
there is	v	hay (presente del verbo "there be" usado delante de formas singulares)	
zero	num	cero	

REPASO 7 (actividad opcional)

Traducir el siguiente fragmento e identificar cada caso de -ing.

1. *User Input*⁵ is an important aspect of programming concepts. This article will teach the basics of user input in the BASIC Programming Language. Please note that the following code may vary from compiler to compiler.

2. Pascal is an influential computer programming language named after the mathematician Blaise Pascal. It was invented by Niklaus Wirth in 1968 as a research project into the nascent field of compiler theory. The acronym PASCAL standing for *primary algorithmic scientific commercial application language* highlights its suitability for computing tasks in science.

3. Covering these topics will be out of this book's scope. Pascal only assumes there *is* some user interface (i.e. a console) and there *are* external entities (this usually refers to "files").

4. *Identifiers* are names denoting constants, types, variables, procedures, and functions. They must begin with a letter, followed by a combination and numbers of letters and digits. The spelling of an identifier is significant. Corresponding upper-case and lower-case letters are considered equivalent.

5. Supplying concatenated strings to routines such as `write/writeLn`, however, may possibly be disadvantageous: The concatenation, especially of long strings, first requires to allocate enough memory to accommodate for the entire resulting string. Then, all the operands are *copied* in their respective location. Hence, in the case of `write/writeLn` it is advisable (for very long strings) to use their capability of accepting an *infinite* number of (comma-separated) parameters.

⁵ user input (s) contribuciones del o los usuarios, datos de entrada del usuario/ de los usuarios

GLOSARIO DEL REPASO 7

<u>Palabra</u>	<u>Función</u>	<u>Significado</u>	<u>Otra función y significado</u>
accommodate	v.	alojar	
acronym	s.	sigla	
advisable	adj.	aconsejable	
after (+ nombre)		por	
allocate	v.	asignar	
basics	s.	fundamentos	
begin	v.	comenzar	
capability	s	capacidad	
compiler	s	compilador	
concatenation	s.	concatenación	
concatenate	v.	concatenar	
cover	v.	cubrir, abordar	
denote	v	denotar (nombrar si otro sentido)	
digit	s	dígitos	
disadvantageous	adj.	desventajoso	
enough	adj.	suficiente	
entire	adj.	completo, total	
entity	s	entidad	
field	s.	campo	
file	s.	archivo	
follow	v.	seguir	
hence	adv.	así, por lo tanto	
highlight	v.	resaltar	
however	adv	sin embargo	
identifier	s	identificador	
input	s	entrada	
its	adj	su/sus	
location	s.	ubicación	
log into	v.	ingresar, registrarse	
lower-case		minúscula	
name	v.	nombrar	
nascent	adj.	incipiente	
operand	s	operando (un elemento en un lenguaje de programación)	
parameter	s	parámetro	
require	v.	requerir	
research	s.	investigación	
respective	adj	respectivo	
some	adj.	algunos/as	
science	s	ciencia	
scope	s.	alcance	
store	v.	almacenar	
significant	adj.	importante, significativo	
suitability	s.	conveniencia, idoneidad	
spell	v.	deletrear	
spelling	s	ortografía	
string	s.	cadena	
such as	adv	tal como	
supply	v.	suministrar	
task	s	tarea	
then	adv	luego, entonces	
teach	v.	enseñar	
upper-case	s	mayúscula	
user input	s	contribuciones de los usuarios, datos de entrada de usuario	
vary	v.	variar	
very	adv.	muy	

EJERCICIO 19.1

Extract from The Ruby on Rails Tutorials

What is Ruby on Rails?

Ruby on Rails is a web framework that was born in the Basecamp company. Characterized by simplicity, it uses the Ruby programming language and is used to build web applications on the server side (back end). It works with the **MVC (Model View Controller⁶)**, a fairly famous architecture pattern.

It was released as open source and was a complete success. At that time, there was no such tool for web development. Its ultimate goal is to deliver the value that the customer or the product needs as quickly as possible.

This technology follows some development principles, such as **Don't repeat yourself** to avoid repeating code. It uses **Convention over Configuration**; this is a software programming paradigm that seeks to minimize the number of decisions a developer needs to make with simplicity and flexibility. In short, it saves you from making decisions, a benefit of Ruby on Rails' agility.

Some examples of companies that use this framework are: Shopify, Github, Airbnb, Groupon, Twitter (nowadays it uses other technologies), AskFM, Slideshare, Hulu, Soundcloud, Urban Dictionary, Basecamp, Bloomberg, Diaspora, Twitch and Heroku.

Features

- It allows you to create RAD⁷ (Rapid Application Development) applications.
- It has a simple and friendly syntax.
- It uses the MVC pattern.
- It supports multiple data bases.
- It uses Active Record, a development pattern to work with databases in a simple way.

Use cases

It is widely used in startups worldwide, because it is designed for agility, rapid development and time saving by avoiding complex architectures.

1.1.2 Conventions in this book

The conventions in this book are mostly self-explanatory. In this section, I'll mention some that may not be. Many examples in this book use command-line commands. For simplicity, all command line examples use a Unix-style command line prompt (a dollar sign), as follows

```
$ echo "hello, world"
hello, world
```

I recommend that users of all operating systems (especially Windows) use a cloud development environment. This is particularly useful because Rails comes with commands which run at the command line. For example, in [Section 1.3.2](#) we'll run a local development web server with the **rails server** command

```
$ rails server
```

As with the command-line prompt, the *Rails Tutorial* uses the Unix convention for directory separators (i.e.,⁸ a forward slash /). For example, the sample application **production.rb** configuration file appears as follows:

```
config/ environments/ productions.rb
```

⁶ Modelo-vista-controlador (es un patrón de arquitectura de software)

⁷ RAD está basado en un modelo iterativo (repetitivo) y para prototipos con ningún (o menos) planeamiento específico.

⁸ i.e., = id est = that is (es decir)

EJERCICIO 19.2

Califique los siguientes enunciados como verdadero (V) o falso (F) según el texto 19.1. Traduzca las verdaderas y justifique las falsas.

1) Ruby on Rails is used for building web applications.

.....

2) When Ruby on rails appeared, it was not the first programming language of this type.

.....

3) It is not one of a simple, fast tool for web development.

.....

4) Famous companies, such as Shopify, Twitter, Bloomberg, still use this framework.

.....

5) Startups use other frameworks as a first option.

.....

6) It is advisable to use a cloud environment in all operating systems.

.....

GLOSARIO DEL EJERCICIO 19.1 (Extract from The Ruby on Rails Tutorials)

<u>Palabra</u>	<u>Función</u>	<u>Significado</u>	Otra función y significado
all	adj	todos/as	
appear	v	aparecer	
as	adv	como	
at	prep	en	
avoid	v	evitar	
back end	s	código del programa	
because	conj	porque	
built-in	adj	integrado/a	
characterize	v	caracterizar	
cloud	s	nube	
development	s	desarrollo	
directory	s	directorio	
environment	s	medio, medioambiente	
fairly	adv	bastante	
file	s	archive	
flexibility	s	flexibilidad	
follow	v	seguir	
for	prep	para	
forward	adj	inclinada, hacia adelante	
framework	s	estructura, soporte	
i.e.	adv	es decir (Latin, abbrev = id est = that is)	
in short	adv	en resumen	
its	adj	su/sus	
many	adj	muchos/as	
mostly	adv	principalmente	
operating	adj	operativo	
paradigm	s	paradigma	
prompt	s.	carácter/es	
run	v	ejecutar	
save	v	ahorrar	
saving	s	ahorro	
section	s	sección	
seek	v	buscar	
self-explanatory	adj	auto-explicativo	
separator	s	separador	
server side	s	servidor	
sign	s	signo	
simplicity	s	simplicidad	
slash	s	barra	
some	adj	algunos/as	
such as	adv	tal como	
success	s	éxito	
startup	s	empresa de nueva creación con un modelo de negocio escalable y uso de las TIC	
syntax	s	sintaxis	
that	conj	que	
this	adj	este/esta	
which	pron	que, el cual/la cual	
with	prep	con	
worldwide	adj	mundial	
yourself	pron	vos/Ud. mismo	

LOS PRONOMBRES OBJETIVOS

EJERCICIO 20.1

Reemplace las palabras subrayadas por los pronombres correspondientes y traduzca las oraciones con dichas substituciones, subrayando los correspondientes pronombres (en inglés y en castellano).

1) A chemical engineer is often concerned with the characteristics of mixtures and solutions.

.....

2) These separations involve changes in that composition.

.....

3) Few students can make use of the same principle of mass transfer.

.....

4) Dr. Jones and I were studying processes to separate or purify chemicals by distillation

.....

5) Mary gave Jim several assignments in advance.

.....

EJERCICIO 20.2

Traduzca las siguientes oraciones

1) You must show us the way to design it.

.....

2) They don't know you but they know me very well.

.....

3) Filtration and settling are required to segregate them.

.....

4) I didn't believe him until I saw you.

.....

5) The screen was behind her.

.....

6) They told us that it was not for us.

.....

7) Slide it in front of them.

.....

8) Let's keep them for him.

.....

9) He can guide you through them.

.....

10) Please, send it to me by plane.

.....

LOS ADJETIVOS POSESIVOS

EJERCICIO 21.1

Complete los espacios en blanco con el ADJETIVO POSESIVO correspondiente al sujeto de cada oración y traduzca. Subraye el adjetivo posesivo en castellano.

1) The gas in the container was not at _____ initial pressure and temperature.

.....

2) Nitrogen and oxygen will mix until _____ concentration is uniform throughout.

.....

3) Charles supposed that _____ mixture had only two components present.

.....

4) We may visualize _____ process by considering Fig. 12.

.....

5) I brought _____ pressure gage for the experiment.

.....

6) The vessel and _____ boiling water will be placed on that table.

.....

7) Those students had _____ class in the new laboratory.

.....

8) Mrs. Douglas left _____ books here because we needed them.

.....

9) Note that _____ results will not always be the same.

.....

10) Gas diffusion and _____ concentration gradient are referred to mathematically.

.....

GLOSARIO DE LOS EJERCICIOS 20.1 y 20.2

<u>Palabra</u>	<u>Función</u>	<u>Significado</u>	Otra función y significado
believe	v	creer	
characteristic	s	característica	
chemical	adj	químico	
composition	s	composición	
concern	v	relacionar, ocuparse, preocuparse	
distillation	s	destilación	
filtration	s	filtrado, filtración	
her	pron obj	la, le, ella	
him	pron obj	lo, le, el	
involve	v	implicar, involucrar, incluir	
me	pron obj	me, mi	
mixture	s	mezcla	
please	expr	por favor	
principle	s	principio	
purify	v	purificar	
segregate	v	segregar	
send	v	enviar (irreg. sent, sent)	
separate	v	separar	
separation	s	separación	
settling	s	decantación, asentamiento	
solution	s	solución	
them	pron obj	los, las, les, ellos, ellas	
transfer	s	transferencia	v transferir
us	pron obj	nos, nosotros, nosotras	

GLOSARIO DEL EJERCICIO 21.1

<u>Palabra</u>	<u>Función</u>	<u>Significado</u>	Otra función y significado
always	adv	siempre	
boil	v	hervir	
book	s	libro	
component	s	componente	
container	s	contenedor, recipiente	
diffusion	s	difusión	
gage	s	medidor (pressure gage = manómetro)	
gradient	s	gradiente	
here	adv	aquí, acá	
initial	adj	inicial	s inicial
mathematical	adj	matemático (mathematically = matemáticamente)	
mix	v	mezclar	
Mrs.	s	señora (abrev de Mistress = señora)	
nitrogen	s	nitrógeno	
note	v	notar, observar	s nota
only	adv	solo, solamente	
oxygen	s	oxígeno	
present	adj	presente	s presente
refer (to)	v	referir/se, mencionar	
suppose	v	suponer	
table	s	mesa, tabla (numérica)	
throughout	adv	completamente	prep por todo, en todo
uniform	s	uniforme	
vessel	s	recipiente, vasija, navío	
visualize	v	visualizar	

Chemical & Thermal Processes

INTRODUCTION

In this chapter, we will briefly describe some of the chemical and thermal concepts used in the practice of engineering and show typical examples of their applications. Much of the material contained in this chapter often falls in the areas of chemical and/or mechanical engineering. The application of these principles can be found in such diverse areas as air pollution control, power generation, heat engine design, oil refining, and manufacturing.

Law of Conservation of Mass

One scientific law that provides the foundation for understanding in chemistry is the **law of conservation of matter**. It simply states that "in a chemical reaction, matter is neither created nor destroyed.", that is, the amount of matter in the system stays constant. A concise way of expressing this law is that the amount of matter in a system is *conserved*.

The law of conservation of mass is also known as the "**law of indestructibility of matter**". There is, of course, an exception to the previous statement. In atomic fission, the simple material balance does not hold, since in this process, matter is transformed into energy.



EXAMPLE: Heating 10 gr of CaCO_3 ⁹ produces 4.4 gr of CO_2 ¹⁰ and 5.6 g of CaO ¹¹

Solution

- Mass of the reactants, CaCO_3 : 10g¹⁰
- Mass of the products, CO_2 and CaO : 4.4g + 5.6g = 10g.

A sample of calcium carbonate (CaCO_3). (Public Domain; [Walkerma](#)).

In any chemical change, one or more initial substances change into a different substance or substances. **Both** the initial **and** final substances are composed of atoms because all matter is composed of atoms. According to the law of conservation of matter, we must have the same number and kind of atoms after the chemical change as before the chemical change.

It may seem that burning destroys matter, but the same amount of matter, or mass, still exists after a fire as before. Figure 5.1.1 shows that when wood burns, it combines with oxygen and changes not only to ash, but also to carbon dioxide and water vapor. The gases float off into the air, leaving behind just the ashes. Suppose we measure the mass of the wood before it burns and the mass of the ashes after burning. Also suppose we measure the oxygen used by the fire and the gases produced by the fire. What will we find? The total mass of matter after the fire will be the same as the total mass of matter before the fire.

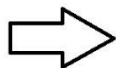


Figure 5.1.1

Burning is a chemical process. The flames are caused as a result of a fuel undergoing combustion (burning).

⁹ Carbonato de calcio

¹⁰ Dióxido de carbono

¹¹ Oxido de calcio

For an example of the application of a material balance, let us consider a power plant. We are interested in calculating the quantity and composition of the stack gas, i.e., in particular how much sulfur dioxide and particulate matter is produced by a typical power plant. Let us assume, for purposes of this example, that we will burn 560,000 pounds of coal per hour (this is *approximately* the amount required for *one* 800-megawatt power plant).

In the combustion process, the coal is burned in the presence of air, liberating heat and producing combustion products. Coal consists chemically of carbon, hydrogen, oxygen, nitrogen and inerts (ash). During combustion, the carbon is converted to either carbon dioxide or carbon monoxide and the hydrogen to water vapor. Essentially, all of the carbon (more than 99.9%) is converted to CO₂ and very little to CO. As an example of our material balance calculation, consider the coal described in the following table:

Constituent	Weight percent
Carbon	77
Hydrogen	-5
Sulfur	8
Water	3
Ash	7

Let us suppose that the combustion takes place with 20% excess air (20% more oxygen than required for burning all the combustibles). Power plant furnaces seldom run at very low or zero excess air because of the tendency for smoke formation. Typical values of excess air are 10-20% for fuel oil and 20-40% for coal. We wish to calculate the amount and percentage of CO₂, SO₂, and ash in the combustion gases when 560,000 lb/hr. of coal are burned. First, we must calculate the number of pounds of carbon, hydrogen, and sulfur that are burned each hour, using the analyses above and the coal firing rate (560,000 lb/hr.).

$$\text{Weight carbon burned} = \left(0.77 \frac{\text{lb carbon}}{\text{lb coal}}\right) \left(560,000 \frac{\text{lb coal}}{\text{hr}}\right) = 431,000 \text{ lbs/hr.}$$

$$\text{Weight hydrogen burned} = \left(0.05 \frac{\text{lb hydrogen}}{\text{lb coal}}\right) \left(560,000 \frac{\text{lb coal}}{\text{hr}}\right) = 28,000 \text{ lbs/hr.}$$

$$\text{Weight sulfur burned} = \left(0.08 \frac{\text{lb sulfur}}{\text{lb coal}}\right) \left(560,000 \frac{\text{lb coal}}{\text{hr}}\right) = 44,800 \text{ lbs/hr.}$$

Now, we must calculate the number of moles of these components burned in order to calculate the air requirements (using the molecular weights of the species involved).

$$\text{Moles carbon burned} = \frac{431,000 \text{ lb carbon hr}}{12 \text{ lbs/lb mole}} \text{ (molecular weight of carbon is 12)} = 35,900 \text{ lb moles/hr}$$

$$\text{Moles hydrogen (H}_2\text{) burned} = \frac{28,000 \text{ lb/hr}}{2 \text{ lbs/lb mole}} \text{ (molecular weight of hydrogen is 2)} = 14,000 \text{ lbs mole/hr}$$

$$\text{Moles sulfur burned} = \frac{44,800 \text{ lb/hr}}{32 \text{ lbs/lb mole}} \text{ (molecular weight of sulfur is 32)} = 1,400 \text{ lbs mole/hr}$$

The theoretical amount of oxygen required to burn each of these components is calculated as follows:

For carbon, which burns to form CO₂, one mole of oxygen is required for each mole of carbon burned. Hence, 35,900 lb moles of carbon require 35,900 lb moles of oxygen. Furthermore, each lb mole of any gas occupies 379 ft³ measured (at 14.7 psia and 60° F); thus, for burning carbon, the oxygen requirement is

$$\text{O}_2 \text{ required for carbon} = (35,900 \text{ lb moles/hs}) \left(379 \frac{\text{ft}^3}{\text{lb mole}}\right) = 13,600,000 \text{ ft}^3/\text{hr}$$

Similarly, for hydrogen, which burns to form water vapor, one -half mole of oxygen is required for each mole of hydrogen burned. Hence, 14,000lb moles of hydrogen require 0.5 (14,000) = 7,000 lb moles of oxygen. In cubic feet,

$$\text{O}_2 \text{ required for hydrogen} = (7,000 \text{ lb moles/hs}) \left(379 \frac{\text{ft}^3}{\text{lb mole}}\right) = 2,650,000 \text{ ft}^3/\text{hr}$$

For combustion of sulfur, which burns to form SO₂, one mole of oxygen is required for each mole of sulfur burned. Hence, 1,400 lb moles of sulfur require 1,400 lb moles of oxygen. In cubic feet,

$$\text{O}_2 \text{ required for sulfur} = (1,400 \text{ lb moles/hr}) \left(379 \frac{\text{ft}^3}{\text{lb mole}} \right) = 530,000 \text{ ft}^3/\text{hr}$$

The theoretical O₂ required to burn carbon, hydrogen and sulfur, then is

$$\text{Theoretical O}_2 \text{ requirement} = 13,600,000 + 2,650,000 + 530,000 = 16,780,000 \text{ ft}^3/\text{hr}$$

Since air contains only 21% oxygen, then the amount of air required to provide 16,780,000 ft³/hr of oxygen is

$$\text{Theoretical air requirement} = \frac{16,780,000 \text{ ft}^3 \text{ O}_2/\text{hr}}{0.21 \text{ ft}^3 \text{ O}_2/\text{ft}^3 \text{ air}} = 79,700,000 \text{ ft}^3 \text{ air/hr}$$

However, this furnace will operate with 20% excess air. Hence, the actual air requirement is the theoretical air requirement plus 20% of the theoretical air requirement.

$$\text{Actual air requirement} = 1.20 (79,700,000 \text{ ft}^3) = 95,600,000 \text{ ft}^3/\text{hr}$$

Note that nearly 100 million ft³/hr is required to burn the coal for an 800-megawatt power plant.

EJERCICIO 22.2

Extraiga del texto anterior la siguiente información:

1) What does the Law of Conservation of matter state?

.....

2) Why cannot the simple material balance be applied in atomic fission?

.....

3) What happens with the number of atoms and the amount of mass before and after a chemical change?

.....

2) Definition of “excess air” in a combustion process.

.....

3) Results of running a power plant with zero excess air.

.....

5) Moles of oxygen required for burning 5.5 moles of carbon.

.....

6) An example of the approximate weight of ash that may be found after burning 5000 lb of coal.

.....

7) Differences in the amounts of air needed for the combustion of fuel oil and coal.

.....

8) Fields of engineering related to the subjects contained in the text.

.....

9) An approximate idea of the coal consumption of a 400-megawatt power plant (in kg/hr).

.....

10) The percentage of carbon that may usually be converted into carbon monoxide when coal is burned in a power plant.

.....

GLOSARIO DEL EJERCICIO 22.1 - Chemical & Thermal processes

<u>Palabra</u>	<u>Función</u>	<u>Significado</u>	<u>Otra función y significado</u>	
actual	adj	real		
amount	s	cantidad		
analysis	s	análisis		
as	adv	como		
assume	v	suponer		
atomic	adj	atómico		
balance	s	equilibrio, balance	v	equilibrar, balancear
before	prep	antes (de)		
briefly	adv	brevemente		
both... and...	exp	tanto... como...		
burn	v	quemar		
calculate	v	calcular, estimar		
call	v	llamar/se	s	llamada
certain	adj	cierto/a/s		
consist	v	consistir		
contain	v	contener		
concise	adj	conciso		
conserve	v	conservar		
convert	v	convertir		
cubic	adj	cúbico		
diverse	adj	diverso		
essentially	adv	esencialmente		
fall	v	caer/se	s	caída, catarata, otoño
firing rate	expr	poder calorífico		
flame	s	llama, fuego		
float off	v	disiparse		
form	v	formar	s	forma
foundation	s	base/s		
fuel oil	s	combustibles líquidos		
furthermore	adv	además		
highest	adj comparativo	más alto		
hence	adv	así, luego, por lo tanto		
hold	v	mantener, fijar (irreg. held, held)		
how much	adv	cuanto		
indestructibility	s	indestructibilidad		
inert	s	materiales inertes		
kind	s	clase, tipo		
knowledge	s	conocimiento		
laid	v	sentó - <i>pasado de lay (colocar, sentar)</i>		
liberate	v	liberar		
little	adv	poco		
many	adj	muchos/as		
mole	s	mol		
nearly	adv	casi		
neither	adv	ni		
nor	prep	ni		
note	v	observar		
occupy	v	ocupar		
pound	s	libra (450 gr); abrev lb		
psia	abrev	pounds per square inch absolute: libras por pulgadas cuadrada absoluta		
quantity	s	cantidad		
react	v	reaccionar		
reactant	s	reactivo		
seem	v	parecer		
state	v	enunciar, asegurar		
statement	s	enunciado		

seldom	adv	rara vez		
simply	adv	simplemente		
since	conj	dado que, ya que	prep	desde
smoke	s	humo		
species	s	elemento, especie, clase		
stay	v	permanecer		
such	adj	tales, dichos. En singular: such a = tal, dicho		
sulfur	s	azufre		
take place	v (expr)	tener lugar, ocurrir		
then	adv	luego, entonces		
theoretical	adj	teórico		
therefore	adv	por lo tanto		
times	s	veces		
under	prep	bajo		
weight	s	peso, pesa		
which	pron relativo	que, el/la cual, los/las cuales	pron interr	cuál
why	pron	por qué		
wish	v	desear	s	deseo
wood	s	madera		

EJERCICIO 23.1

Lea el siguiente texto

Direct current electricity

All circuits must contain a source of electromotive force to establish the difference of potential that makes possible the current flow. This source may be a dry cell, a mechanical generator, or any of the other devices that will be discussed later in this book. All paths of the circuit lead in closed loops from the high-potential (negative) end of the electromotive-force source to its low-potential (positive) end.

The current in a circuit may flow through solid conductors, liquids, gases, vacuums, or any combinations of these. Its path may include lamps, toasters, motors, or any of the thousand-and-one electrical devices available in this electrical age. But, regardless of the type of circuit and the devices through which the current flows, all circuits offer some resistance to the current.

This resistance may be high or low, depending upon the type of circuit and the devices employed. Sometimes this resistance is undesirable, as, for example, the resistance of the wires connecting the various devices in a circuit. Accordingly, we keep this resistance at a low level by using wires made of copper having a large cross-sectional area, and by keeping their lengths as short as possible. Sometimes, however, it is desirable to introduce concentrated, or lumped, resistances into the circuit. Such a lumped resistance is called a **resistor**. In electrical diagrams, the symbol for a **fixed** resistor – that is, one whose resistance is constant- is 

The symbol for a **variable** resistor is 

In this chapter we will consider three general types of circuits. One is the **series** circuit, which offers a single, continuous, external path for current flow from the negative side of the electromotive-force source to the positive side. Another is the **parallel** circuit, which offers two or more parallel paths for current flow from negative to positive. The third type is the **series-parallel** circuit, a combination of the other two.

EJERCICIO 23.2

Traduzca las siguientes oraciones y decida si son **verdaderas** o **falsas** de acuerdo al texto en EJERCICIO 23.1

1) A resistor is a lumped resistance.

.....

2) They use copper wires with a large cross-sectional area in order to keep resistance low.

.....

3) There are a thousand and one electrical devices in this electrical age.

.....

4) Most circuits offer some resistance to the current.

.....

5) The parallel circuit offers a parallel path for the current flow.

.....

GLOSARIO DEL EJERCICIO 23.1

<u>Palabra</u>	<u>Función</u>	<u>Significado</u>	Otra función y significado
accordingly	adv.	así, en consecuencia	
age	s.	era, edad	
electromotive	adj.	electromotriz	
establish	v.	establecer	
fixed	adj.	fijo	
lead	v.	conducir	s. plomo
loop	s.	vuelta, lazo, bucle	
lump	v.	agrupar	
parallel	adj.	paralelo	
path	s.	camino, recorrido, trayectoria	
toaster	s.	tostadora	
undesirable	adj.	indeseable	
vacuum	s.	vacío	

EL ADVERBIO

EJERCICIO 24.1

Complete los espacios con los adjetivos dados entre paréntesis o con los adverbios correspondientes y traduzca.

1) We will _____ describe some of the chemical and thermal concepts used in the practice of engineering. (brief)

.....
.....

2) How much sulfur dioxide is produced by a _____ power plant? (typical)

.....
.....

3) This is _____ the amount required for one 800-megawatt power plant. (approximate)

.....

4) Coal consists _____ of carbon, hydrogen, oxygen, nitrogen and inerts. (chemical)

.....
.....

5) It was a _____ defined concept of what is understood by "excess air". (good)

.....

6) He hardly speaks English because he finds it very _____. (hard)

.....

7) In that type of circuit, the current flow will be _____ small. (considerable)

.....
.....

8) The law of conservation of matter states _____ (simple) that matter cannot be created or destroyed.

.....
.....

9) (Essential) _____ all of the carbon is converted to CO₂, and very little to CO.

.....

10) It is (actual) _____ very simple to learn how to print text.

.....

NUMEROS ORDINALES Y CARDINALES

1 one	1st first
2 two	2nd second
3 three	3rd third
4 four	4th fourth
5 five	5th fifth
6 six	6th sixth
7 seven	7th seventh
8 eight	8th eighth
9 nine	9th ninth
10 ten	10th tenth
11 eleven	11th eleventh
12 twelve	12th twelfth
13 thirteen	13th thirteenth
14 fourteen	14th fourteenth
15 fifteen	15th fifteenth
16 sixteen	16th sixteenth
17 seventeen	17th seventeenth
18 eighteen	18th eighteenth
19 nineteen	19th nineteenth
20 twenty	20th twentieth
21 twenty-one	21st twenty-first
22 twenty-two	22nd twenty-second
30 thirty	30th thirtieth
40 forty	40th fortieth
50 fifty	50th fiftieth
60 sixty	60th sixtieth
70 seventy	70th seventieth
80 eighty	80th eightieth
90 ninety	90th ninetieth
100 a/one hundred	100th hundredth
101 a/one hundred and one	101st hundred and first
200 two hundred	200th two hundredth
1,000 a/one thousand	1,000th thousandth
10,000 ten thousand	10,000th ten thousandth
100,000 a/one hundred thousand	100,000th one hundred thousandth
1,000,000 a/one million	1,000,000th one millionth
1,000,000,000 a/one billion (en castellano 10^{12})	1,000,000,000th one billionth

Nota: Observe que los miles se separan con comas

EJEMPLOS:

1,005	one thousand and five
1,608	one thousand six hundred and eight
90,435	ninety thousand four hundred and thirty-five
500,000	five hundred thousand
3,755,372	three million seven hundred and fifty-five thousand three hundred and seventy-two

DECIMALES

Se escribe	Se expresa:
0.3	nought point three zero point three
3.45	three point four five (NOT three point forty-five)
98.4	ninety-eight point four
\$1.55	one dollar, fifty-five cents one dollar, fifty-five
\$700.00	seven hundred dollars
€3,500.50	three thousand five hundred euro and fifty cents three thousand five hundred euro, fifty cents

Nota: Observe que los decimales se separan con punto

EJEMPLOS:

0 zero (también se lee como la letra “o”)
 0.5 = .5 zero point five (point five)
 0.01 = .01 zero point “o” one (point “o” one)
 0.001 = .001 zero point double “o” one (point double “o” one)
 3.1416 three point one four one six

FRACCIONES Y OTRAS EXPRESIONES MATEMÁTICAS

Se escribe:	Se expresa
$\frac{1}{2}$	a half ó one half
$\frac{1}{4}$	a quarter ó one quarter
$\frac{3}{4}$	three quarters
$\frac{1}{3}$	a third ó one third
$\frac{2}{3}$	two thirds
$\frac{1}{5}$	a fifth ó one fifth
$\frac{3}{5}$	three fifths
$\frac{1}{8}$	an eighth ó one eighth
$\frac{5}{8}$	five eighths
$1\frac{1}{2}$	one and a half
$5\frac{3}{4}$	five and three quarters
$3\frac{5}{12}$	three and five twelfths
$\frac{1}{n}$	one nth
$X = a^2$	X equals a squared
$X = a^3$	X equals a cubed
a^n	a to nth (power)
a^{-n}	a to the minus nth
$\sqrt{a}$	square root of a
$\sqrt[3]{a}$	cubed root of a

OPERACIONES MATEMÁTICAS

Operation	Verbo	Se expresa
Addition	To add	$a + b = c$: a plus b equals c
Subtraction	To subtract	$a - b = c$: a minus b equals c
Multiplication	To multiply	$a \times b = c$: a times b equals c
Division	To divide	$a / b = c$: a divided by b equals c
Differentiation	To differentiate	dx: differential dx
Derivation	To derive	dy/dx: derivative of y with respect to x dx/dt: time derivative of x
Integration	To integrate	$\int_a^b f(x) dx$: integral of x from a to b of f of x x dx

OTRAS EXPRESIONES USADAS EN MATEMÁTICAS O GEOMETRÍA

digit	dígito
figure	cifra
integer	entero
a is greater than b	$a > b$
a is less than b	$a < b$
the ratio of a to b	el cociente entre a y b (cociente de a sobre b)
at random	al azar
arithmetic mean	media aritmética
straight line	línea recta
locus	lugar geométrico (o representación gráfica=
slope	pendiente
to label the axes	marcar/rotular los ejes
to plot y against x	representar (graficar) y en función de x
full/solid line	línea de trazo continuo
dashed line	línea de trazos
dotted line	línea de puntos
cutaway view	vista de corte
scale	escala
scale model	modelo a escala
full size	escala natural/real
80 km an hour	80 km por hora
once	una vez
twice	dos veces
three times	tres veces

EJERCICIO 25.1

Escriba en números las siguientes expresiones.

- 1) two fifths plus three and a half
- 2) square root of nine hundred thousand and eighty-one
- 3) thirteen hundred times eight hundred and seventy-eight
- 4) point o three squared plus nineteen cubed
- 5) four thousand and six times point two five
- 6) ten halves minus x to the nth
- 7) seventeen hundred point four nine divided by two to the eleventh
- 8) the fourth root of forty million two hundred and sixty-one thousand four hundred and one:
- 9) two sevenths divided by the square root of seventy-nine thousand five hundred
- 10) thirteen squared minus thirty cubed

GLOSARIO GENERAL

<u>Palabra</u>	<u>Función</u>	<u>Significado</u>	<u>Otra función y significado</u>	
A				
a	art	un, uno, una		
ability	s	habilidad, capacidad		
about	prep.	sobre, acerca de		
above	prep	arriba, por encima de		
access	v	acceder		
accommodate	v.	alojar		
according to	adv	de acuerdo con, según		
accordingly	adv	de esa manera, en consecuencia		
acronym	s.	sigla		
actual	adj	real		
adjust	v.	ajustar		
admittance	s	admitancia		
advance	s	adelanto, avance, anticipación	v	adelantar
			in advance = con anticipación/por adelantado	
advisable	adj.	aconsejable		
affect	v.	afectar		
after (+ nombre)		por		
actual	adj	verdadero, real		
add	v	agregar, sumar		
against	prep	contra		
age	s.	era, edad		
air	s	aire		
aircraft	s	aeronave		
algebraic	adj	algebraico		
all	adj	todos/as		
allocate	v.	asignar		
allow	v.	permitir		
alloy	s	aleación		
almost	adv	casi		
always	adv	siempre		
am	v	soy, estoy		
amount	s	cantidad	v	valer
an	art	un, uno, una		
analysis	s	análisis		
and	conj	y		
annual	adj	annual		
anode	s	ánodo		
any	adj	cualquier (afir), algún (interr), ningún (neg)		
anyone	adj	cualquiera (afir), alguien (interr), nadie (neg)		
anything	adj	cualquier cosa (afir), algo (interr), nada (neg)		
appear	v	aparecer		
application	s	aplicación, solicitud		
apply	v	aplicar		
appropriate	adj	apropiado		
approximately	adv	aproximadamente		
are	v	eres, sos, son, estás, están		
area	s	area		
argument	s.	argumento		
array	s.	arreglo, matriz		
as	adv	como	conj. = como	adv. = a medida que
as... adj ... as	exp.	tan... (adj) como		
ash	s	ceniza		
ask for	v.	pedir		

as regards	frase adv.	con respecto a
assignment	s	tarea (asignada)
assume	v	suponer
at	prep	en, a
atomic	adj	atómico
attempt	s	intento
auxiliary	adj	auxiliar
available	adj	disponible
average	s	promedio, medio
avoid	v	evitar
away	adv	afuera, dirección contraria
axis	s	eje

B

bacillus	s.	bacteria	
B.C.	abrev	antes de Cristo (Before Christ)	
back end	s	código del programa	
bad	adj	malo	
bail	s	barra de sujeción	
balance	s	equilibrio, balance	v equilibrar, balancear
bar	s	barra, bar	
based	adj.	basado	
basic	adj	básico	
basics	s.	fundamentos	
basis	s	fundamento, base	
battery	s	bacteria	
beam	s	haz, rayo, viga	
because	conj	porque	
become	v	hacerse, volverse, convertirse (irreg. became, become)	
before	prep	antes de, delante de	
begin	v	empezar (irreg. began, begun)	
behavior	s	comportamiento	
behind	prep	detrás de	
believe	v	creer	
below	prep	por debajo de	
beside	prep	al lado de, junto a	
between	prep	entre	
beyond	prep	más allá	
boil	v	hervir	
book	s	libro	
both	adj	ambos	
both... and...	exp	tanto... como...	
box	s	caja, cajón	
brackets	s.	corchetes	
branch	s	rama, sucursal	
break	v	romper, quebrar (irreg. broke, broken)	break up = separar
break	s	pausa, descanso	
briefly	adv	brevemente	
bring	v	traer (irreg. brought, brought)	
broadcasting	s	transmisión	
building	s.	sustantivo	
built-in	adj	integrado/a	
burn	v	quemar	
bus	s	ómnibus	
but	conj,	pero, sino	prep excepto
buzz	s	zumbido, chicharra	
by	prep	por	(by bus = en ómnibus)

C

calculate	v	calcular		
calculation	s	cálculo		
call	v	llamar/se	s	llamada
can	v. modal	poder		
capability	s	capacidad		
capita	s	cápita, cabeza		
car	s	auto		
carbon	s	carbón		
careful	adj	cuidadoso		
carefully	adv	cuidadosamente		
carrier	s	portador/a		
case	s	caso		
cause	v	causar, hacer (que)		
century	s	siglo		
certain	adj	cierto/a/s		
challenge	s.	desafío		
change	v	cambiar, variar	s	cambio
chapter	s	capítulo		
character	s	carácter (letra)		
characteristic	s	característica		
characterize	v	caracterizar		
charge	s	carga	v	cargar
cheap	adj	barato		
chemical	adj	químico		
chief	s	jefe	adj	principal
child	s	niño, chico/a		
choice	s	elección		
choose	v	elegir (irreg. chose, chosen)		
circuit	s	circuito		
class	s	clase		
clear	adj	claro		
close	adj	cercano (close to =cercano a)	v	cerrar
cloud	s	nube		
coal	s	carbón (mineral)		
colon	s	dos puntos (:)		
color	v	colorear	s	color
column	s	columna		
combustion	s	combustión		
comfortable	adj	confortable		
comma	s	coma (,)		
command prompt	s	línea de comandos, símbolo del sistema		
command	s	comando, orden	v	ordenar
comment	s	comentario	v	comentar
compiler	s	compilador		
complete	v	completar	adj	completo
complexity	s	complejidad		
component	s	componente		
computer	s	computadora		
concatenation	s.	concatenación		
concatenate	v.	concatenar		
concept	s	concepto		
concern	v	relacionar, ocuparse, preocuparse		
conclude	v	concluir		
condition	s	condición		
conservation	s	conservación		
consider	v	considerar		
consist	v	consistir		

contain	v	contener		
concise	adj	conciso		
conserve	v	conservar		
constant	adj	constante	s	constante
consumption	s	consumo		
contain	v	contener		
container	s	contenedor, recipiente		
control	s	control		
convention	s	convención, uso		
conversion	s	conversion		
convert	v	convertir/se		
convince	v.	convencer		
cool	v.	refrigerar		
copy	s	copia	v	copiar
count	v	contar		
country	s	país, campo		
course	s	curso		
cover	v.	cubrir, abordar		
create	v	crear		
creep	s.	deformación por fluencia		
criterion	s	criterio		
critical	adj	crítico		
cross-sectional	adj	transversal		
cubic	adj	cúbico		
current	s	corriente		
curve	s	curva		
cut	v	recortar, reducir		

D

dangerous	adj	peligroso		
datum	s	dato		
decrease	v.	disminuir, decrecer		
define	v	definir		
deformation	s	deformación		
degree	s	grado		
delay	v.	retrasar		
deliver	v	entregar, suministrar		
demand	s	demanda	v	demandar, exigir
denote	v	denotar (nombrar si otro sentido)		
depend	v	depender (depend on/upon= depender de)		
describe	v	describir		
design	s	diseño	v	diseñar
desire	v	desear	s	deseo
destroy	v	destruir		
determine	v	determinar		
develop	v	desarrollar		
development	s	desarrollo		
device	s	aparato, dispositivo, artefacto		
diameter	s	diámetro		
die	s	matriz	v	morir
difference	s	diferencia		
different	adj	diferente		
difficult	adj	difícil		
digit	s	dígitos		
dioxide	s	dióxido, anhídrido		
directory	s	directorío		
disadvantageous	adj.	desventajoso		
discuss	v	discutir, tratar		

dish	s	plato		
disk	s	disco		
disperse	v	dispersar		
display	v	mostrar, exhibir	s	muestra, exhibición
distillation	s	destilación		
distinguish	v	distinguir		
distortion	s	distorsión		
diverse	adj	diverso		
divide	v	dividir		
dollar	s	dólar		
door	s	puerta		
drop	s	caída, gota	v	dejar caer
drum	s	tambor		
during	prep	durante		
dynamo	s	dínamo, generador		

E

e.g.	abrev	por ejemplo (abreviatura de exempli gratia)		
each	adj	cada	pron	cada uno
early	adj	temprano, primitivo, primeros		
earth	s	tierra		
Earth	s	Tierra (planeta)		
easy	adj	fácil		
echo	s	eco, repetición	v	producir eco
edit	v	editar		
effect	s	efecto		
efficiency	s	eficiencia, rendimiento		
either	expr	(indica que se van a enumerar opciones, es posible no traducirla)		
elastically	adv	elásticamente		
electrical	adj	eléctrico		
electricity	s	electricidad		
electromotive	adj.	electromotriz		
element	s	elemento		
elongation	s	alargamiento		
emit	v.	emitir		
emphasis	s	énfasis		
encyclopedia	s	enciclopedia		
end	s	extremo, fin	v	finalizar
energy	s	energía		
enough	adj.	suficiente		
engine	s	motor, máquina		
engineer	s	ingeniero		
English	s	inglés	adj	inglés
environment	s	medio, medioambiente		
environmental	adj	ambiental		
entire	adj.	completo, total		
entity	s	entidad		
equal	adj	igual		
error	s	error		
especially	adv	especialmente		
essentially	adv	esencialmente		
establish	v.	establecer		
even	adv	incluso, aún	adj	parejo, par
every	adj	cada, todo		
example	s	ejemplo		
exceed	v.	exceder		
exception	s	excepción		
excess	s	exceso		

exist	v	existir
expect	v	esperar
expensive	adj	caro
explain	v	explicar
express	v	expresar

F

fabrication	s	fabricación		
fact	s	hecho		
fairly	adv	bastante		
fall	v	caer/se	s	caída, catarata, otoño
fast	adj	rápido		
feedback	s	realimentación		
few	adj	pocos (a few = unos pocos)		
field	s.	campo		
figure	s	figura, cifra	v	figurar, imaginar
file	s	archivo, lima	v	archivar, limar
filing	s	limadura		
filtration	s	filtrado, filtración		
final	adj	final		
find	v	encontrar (irreg. found, found)		
fine	adj	fino		
fire	v	alimentar, disparar	s	fuego
firing rate	expr	poder calorífico		
first	adj	primero		
fission	s	fisión		
fixed	adj.	fijo		
flame	s	llama, fuego		
flash	s	destello	v	destellar
flexibility	s	flexibilidad		
float	v.	flotar		
float off	v	disiparse		
floor	s	piso		
focus	s.	foco		
follow	v	seguir		
food	s	comida, alimento		
foot	s	pie		
for	prep	para, por		
form	v	formar	s	forma
format	s	formato		
formation	s	formación		
forward	adj	inclinada, hacia adelante		
fossil	s	fósil	adj	fósil
four	numer	cuatro		
foundation	s	base/s		
fracture	v	fracturar		
framework	s	estructura, soporte		
frequent	adj	frecuente		
from	prep	de, desde		
fuel	s	combustible		
fuel oil	s	combustibles líquidos		
fulfill	v	cumplir, llenar		
full	adj	lleno, completo		
function	s	función	v	funcionar
furnace	s	horno (industrial), caldera		
furthermore	adv	además		
future	s	future		

G

gage	s	medidor (pressure gage = manómetro)	
gas	s	gas (a veces gasolina)	
generate	v	generar	
generation	s	generación	
geographical	adj	geográfico	
get	v	obtener, conseguir, etc. (irreg. got, got o gotten)	
get implemented	v	implementarse	
give	v	dar (irreg. gave, given)	
glass	s	vidrio (sust. incontable que puede usarse en plural), vaso (sust contable)	
go	v	ir (irreg. went, gone)	
goal	s	objetivo	
go into	v.	destinar	
go on	v.	suceder	
good	adj	bueno	
gradient	s	gradiente	
graphic	s	gráfico	
great	adj	gran, grande	
group	s	grupo	
grow	v	crecer (irreg. grew, grown)	
growth	s	crecimiento	
guide	s	guía	v guiar
guideline	s	pauta, directiva	

H

half	s	mitad	
happen	v	suceder	
hardly	adv.	rara vez, casi, apenas	
have	v	tener (irreg. had, had)	
heat	s	calor	
heat	v.	calendar	
heavy	adj	pesado	
help	v	ayudar	s ayuda
hence	adv.	así, por lo tanto	
her	pron obj	la, le, ella	
here	adv	aquí, acá	
hero	s	héroe	
high	adj	alto	
highest	adj comp	más alto	
highlight	v.	resaltar	
him	pron obj	lo, le, el	
hold	v	mantener, fijar (irreg. held, held)	
hope	v	esperar	
home	s	hogar, casa	
house	s	casa	
housing	s.	carcasa, cubierta	
hour	s	hora	
how much	adv	cuanto	
how	pron interr	cómo	
however	adv	sin embargo	
hydroelectric	adj	hidroeléctrico	
hydropower	s.	energía hidráulica	

I

i.e.	adv	es decir (Latin, abbr: id est: that is)
idea	s	idea

ideal	adj	ideal		
identifier	s	identificador		
indestructibility	s	indestructibilidad		
ignore	v	ignorar		
important	adj	importante		
improvement	s.	mejora		
in	prep	en		
in front of	prep	delante de		
in order to	expr	para (expression de propósito)		
inch	s	pulgada		
include	v	incluir		
incorporate	v	incorporar		
increase	s	aumento	v	aumentar
independent	adj	independiente		
index	s.	índice		
indicate	v	indicar		
industrial	adj	industrial		
industrialization	s	industrialización		
industrialize	v	industrializar		
industry	s	industria		
inert	s	materiales inertes		
information	s	información		
initial	adj	inicial	s	inicial
input	s	entrada		
inside	adv	dentro de, adentro	adj	interior
in short	adv	en resumen		
instruction	s	instrucción		
instrument	s	instrumento		
interest	s	interés		
into	prep	en, dentro de		
involve	v	implicar, involucrar, incluir		
iron	s	hierro		
irrigation	s	irrigación		
is	v	es, está		
its	adj pos	su/s		
itself	adj reflex	en sí, propiamente dicho		

J

jet	s	chorro		
job	s.	trabajo		
keep	v	mantener, conservar (irreg. kept, kept)		
keypad	s	bloque de teclas, teclado		
keyword	s	palabra clave		
knife	s	cuchillo		
knob	s	manija, perilla, picaporte		
know	v	saber, conocer (irreg. knew, known)		

K

keypad	s	bloque de teclas, teclado		
keyword	s	palabra clave		
kind	s	clase, tipo		
knife	s	cuchillo		
knob	s	manija, perilla, picaporte		
know	v	saber, conocer (irreg. knew, known)		
knowledge	s	conocimiento		

L

laboratory	s	laboratorio (abrev. Lab)		
laid	v	sentó - <i>pasado de lay (sentar)</i>		
land	v	aterrizar	s	tierra, territorio
landscape	s.	paisaje		
language	s	idioma, lenguaje		
large	adj	grande		
last	adj	último		
late	adj	tarde		
later	comp	más tarde, después		
law	s	ley		
lead	v.	conducir	s. plomo	
leaf	s	hoja		
learn	v	aprender		
leave	v	salir, dejar (irreg. left, left)		
left	s	izquierda	adj	izquierdo
length	s	longitud		
less	adj.	menos		
letter	s	letra, carta		
lever	s	palanca		
liberate	v	liberar		
life	s	vida		
light	adj	liviano, luminoso	s	luz
light	v.	iluminar		
line	s	línea		
linear	adj	lineal		
list	v	listar	s	lista
little	adv	poco		
living	s	vida		
load	s	carga	v	cargar
location	s.	ubicación		
log into	v.	ingresar, registrarse		
long	adj	largo		
look (at)	v	mirar (a)	s	apariencia
loop	s.	vuelta, lazo, bucle		
low	adj	bajo		
lower	adj	inferior		
lump	v.	agrupar		

M

magnet	s	imán		
main	adj	principal		
maintain	v	mantener		
make sure	expr	asegurarse		
make	v	hacer (irreg. made, made)		
maker	s	fabricante		
make decisions	exp.	tomar decisiones		
man	s	hombre		
manufacture	v	fabricar		
manufacturing	adj	manufacturero		
many	adj	muchos/as		
mark	s	marca, símbolo	v	marcar
mass	s	masa		
material	s	material		
mathematical	adj	matemático (mathematically = matemáticamente)		
matter	s	materia		
maximum	s	máximo		

may	v. modal	puede que		
me	pron obj	me, mi		
measure	v	medir		
mechanical	adj	mecánico		
medium	s	medio		
meet	v	reunir, encontrar, satisfacer (irreg. met, met)		
memory	s	memoria		
mention	v	mencionar		
metal	s	metal		
microwave	s	microonda		
middle	s	mitad	adj	medio, central
minimize	v	reducir, minimizar		
minimum	s	mínimo		
mix	v	mezclar		
mixture	s	mezcla		
mobile	adj	móvil		
mode	s	modo		
modulus	s	módulo		
mole	s	mol		
monoxide	s	monóxido		
more	adj	más	adv	más
mostly	adv	principalmente		
mountainous	adj	montañoso		
move	v.	cambiar		
moving	adj	móvil		
Mrs.	s	señora (abrev de Mistress = señora)		
Mr.	s	señor		
much	adj	mucho		
muscle	s	músculo		
must	v. modal	deber		

N

name	v.	nombrar		
narrow	adj	angosto		
nascent	adj.	incipiente		
nearly	adv	casi		
necessary	adj	necesario		
need	v	necesitar	s	necesidad
neither	adv	ni		
new	adj	nuevo		
next	adj	próximo, siguiente		
nitrogen	s	nitrógeno		
nor	conj.	ni		
normal	adj	normal		
note	v	observar	s	nota
notice	v	observar	s	atención, anuncio
nucleus	s	núcleo		
number	s	número		
numeral	s	numeral		
numeric	adj	numérico		
numerical	adj	numérico		

O

obtain	v	obtener		
occupy	v	ocupar		
occur	v	ocurrir, producir/se		
of	prep	de		

of course	expr	por supuesto	
offset	v.	compensar	
often	adv	a menudo	
oil	s	aceite, petróleo	
old	adj	viejo	
on	prep	sobre, en	
only	adv	solo, solamente	
operand	s	operando (un elemento en un lenguaje de programación)	
operate	v	operar, funcionar, hacer funcionar	
operating	adj	operativo	
or	conj	o	
original	adj	original	
over	prep	sobre	
over	adv.	más de	prep. encima (de)
oxide	s	óxido	
oxygen	s	oxígeno	

P

pair	s	par	
paper	s	papel	
paradigm	s	paradigma	
paragraph	s	párrafo	
parallel	adj.	paralelo	
parameter	s	parámetro	
parentheses	s	paréntesis	
particle	s	partícula	
particular	adj	particular	
particulate	s.	materia en forma de partículas	
pass	s	pase	v pasar, aprobar
passage	s	pasaje	
passive	adj	pasivo	
path	s.	camino, recorrido, trayectoria	
pattern	s	patrón, modelo	
people	s	gente, personas, pueblo	
per	prep	por	
percentage	s	porcentaje	
perfect	adj	perfecto	
permanent	adj	permanente	
petroleum	s	petróleo	
phase	s	fase	
phenomenon	s	fenómeno	
physics	s	física	
piano	s	piano	
picture	s	figura, cuadro, película	
pipe	s	caño, tubo	
pipng	s	cañería, tubería	
place	v	colocar	s lugar
plane	s	plano, avión (airplane)	
please	expr	por favor	
point	s	punto	
pole	s	polo, poste	
pollution	s	polución	
population	s	población	
potato	s	papa	
pound	s	libra (450 gr); abrev lb	
powder	s	polvo	
power	s	potencia, poder	
practice	s	práctica	v practicar

prefer	v	preferir			
presence	s	presencia			
present	adj	presente	s	presente	v presentar
press	v	presionar, oprimir, apretar			
pressure	s	presión			
previous	adj	previo			
principle	s	principio			
print	v	imprimir			
problem	s	problema			
process	s	proceso			
processing	s	procesamiento			
product	s	producto			
production	v	producción			
program	s	programa			
progressive	adj	progresivo			
property	s	propiedad			
proportional	adj	proporcional			
provide	v	proveer			
proton	s	protón			
psia	abrev	pounds per square inch absolute = libras por pulgadas cuadrada absoluta			
pull	v	tirar	s	tiro	
pump	s	bomba	v	bombear	
punctuation	s	puntuación			
purify	v	purificar			
purpose	s	propósito, aplicación			
push	v	empujar	s	empuje	

Q

quality	s.	calidad			
quantity	s	cantidad			
quotation marks		comillas			
quote	s	comilla	v	citar	

R

radio	s	radio			
radius	s	radio (de una circunferencia)			
rail	s	riel			
range	v	oscilar, variar			
rate	s	velocidad, ritmo, tarifa, índice	v	clasificar, tasar	
ratio	s	cociente, relación			
ray	s	rayo			
react	v	reaccionar			
reactant	s	reactivo			
recommend	v	recomendar			
recycle	v.	reciclar			
R&D	s	Research and Development – investigación y Desarrollo (I + D)			
reduce	v.	reducir			
refer (to)	v	referirse, mencionar			
refine	v	refinar			
regard	v.	considerar			
region	s	región			
regulation	s.	regulación, ley			
reject	v	rechazar			
relationship	s	relación			
release	v	liberar			
remain	v	permanecer			
remark	s	acotación, comentario			

respective	adj	respectivo		
require	v	requerir		
requirement	s	requisito		
research	s.	investigación		
reserve	s	reserva	v	reservar
resource	s	recurso		
restrict	v	restringir		
result	s	resultado	v	resultar
return	v	volver	s	retorno
revenue	s.	ganancia		
revolution	s	revolución		
right	s	derecha, derecho	adj	derecho, correcto
roll	v	rodar		
roof	s	techo		
rotating	adj	rotativo		
row	s	fila		
rule	s	regla		
run	v	correr, ejecutar, hacer funcionar (irreg. ran, run)		s. recorrido, tramo

S

same	adj	mismo/a (the same as = lo mismo que)		
sample	s	muestra		
save	v	ahorrar		
saving	s	ahorro		
science	s	ciencia		
scope	s	alcance		
score	s	puntaje	v	tener puntaje
screen	s	pantalla		
section	s	sección		
sector	s	sector		
see	v	ver (irreg. saw, seen)		
seek	v	buscar		
seem	v	parecer		
segregate	v	segregar		
seldom	adv	rara vez		
select	v	seleccionar		
selection	s	selección		
self-explanatory	adj	auto-explicativo		
semicolon	s	punto y coma (;)		
send	v	enviar (irreg. sent, sent)		
separate	v	separar		
separation	s	separación		
separator	s	separador		
server	s	servidor		
server side	s	servidor		
service	s	servicio		
set	s	juego, equipo, aparato, conjunto	v	colocar, fijar
settling	s	decantación, asentamiento		
several	adv	varios		
sheet	s	hoja, lámina, chapa		
shift	v	cambiar, conmutar	s	cambio, turno
short	adj	corto, bajo		
show	v	mostrar (irreg. showed, shown)	s	espectáculo
sign	s	signo, cartel	v	firmar
signal	s	señal		
significant	adj.	importante, significativo		
simple	adj	simple		
simplicity	s	simplicidad		

simply	adv	simplemente		
since	conj	dado que, ya que	prep	desde
single	adj	único		
situation	s	situación		
size	s	tamaño, talla		
sky	s	cielo		
slash	s	barra		
slide	v	deslizar/se (irreg. slid, slid)	s	deslizamiento
slow	adj	lento		
small	adj	pequeño		
smoke	s	humo		
society	s	sociedad		
solar	adj	solar		
solution	s	solución		
some	adj	algunos/as		
sometimes	adv	algunas veces		
source	s	fuelle		
sort	s.	clase, tipo		
space	s	espacio	v	separar
speaking	s	habla		
special	adj	especial		
species	s	elemento, especie, clase		
specific	adj	específico		
specification	s	especificación		
specimen	s	espécimen, probeta		
speculate	v	especular		
spell	v.	deletrear		
spelling	s	ortografía		
spend	v.	gastar		
stack	s	chimenea		
stage	s	etapa		
stand for	v	representar, significar		
standard	adj	normal, común	s	norma
state	v	enunciar, asegurar		
statement	s	enunciado, oración		
station	s	estación (power station = estación eléctrica)		
stay	v	permanecer		
steam	s	vapor (de agua)		
steel	s	acero (stainless steel = acero inoxidable)		
step	s	paso	step by step =	paso a paso
stop	v	parar, detener	s	parada
store	v.	almacenar		
strain	s	deformación	v	estirar, tensar
stress	s	esfuerzo, tensión	v	someter a esfuerzo
string	s	cadena (en computación), cuerda		
structure	s	estructura		
student	s	estudiante		
study	v	estudiar	s	estudio
style	s	estilo		
subject	v	someter	s	sujeto, tema
substitution	s	sustitución		
such	adj	tales, dichos. En singular: such a = tal, dicho		
such as	adv	tal como		
success	s	éxito		
suitable	adj.	apropiado/a/os/as		
suitability	s.	conveniencia, idoneidad		
sulfur	s	azufre		
supply	s	suministro, Fuente	v.	suministrar
support	v	soportar, sostener, respaldar	s	soporte

suppose	v	suponer	
surface	s	superficie	
surround	v	rodear, circundar	
startup	s	empresa de nueva creación con un modelo de negocio escalable y uso de las TIC	
switch	s	interruptor (llave)	v. cambiar
symbol	s	símbolo	
syntax	s	sintaxis	
system	s	sistema	

T

table	s	mesa, table (numérica)	
take place	v (expr)	tener lugar, ocurrir	
take	v	tomar, llevar (irreg. took, taken)	
takeoff	s	despegue (aviación)	
tall	adj	alto	
task	s	tarea	
teach	v.	enseñar	
technique	s	técnica	
television	s	televisión	
tell	v	decir (irreg. told, told)	
temperature	s	temperatura	
tendency	s	tendencia	
tensile	s	tensión, tracción	
term	s	término, plazo	
terrain	s	terreno	
test	s	ensayo, prueba	v ensayar, probar
text	s	texto	
than	conj	que (en comparativos)	
that is	expr	es decir	
that	adj dem	ese, esa, aquel, aquella	pron eso, aquello, que
the	art	el, la, los, las	
them	pron obj	los, las, les, ellos, ellas	
then	adv	luego, entonces	conj entonces, en ese caso
theoretical	adj	teórico	
theory	s	teoría	
there are	v	hay (presente del verbo "there be" usado delante de formas plurales)	
there is	v	hay (presente del verbo "there be" usado delante de formas singulares)	
therefore	adv	por lo tanto	
there was	v	había/hubo (pretérito del verbo "there be" usado delante de formas singulares)	
there were	v	había/hubo (pretérito del verbo "there be" usado delante de formas plurales)	
thermal	adj	térmico	
thermodynamics	s	termodinámica	
these	adj dem	estos, estas	pron éstas, estos
thick	adj	grueso, espeso	
thin	adj	delgado	
this	adj dem	este, esta	pron esto
those	adj dem	esos, esas, aquellos, aquellas	pron aquellos, aquellas
through	prep	a través de	
throughout	adv	completamente	prep por todo, en todo
thus	adv	así	
tie	v	atar, ligar	
time	s	tiempo, hora, vez, momento	
to	prep	a, hacia, para (también se utiliza para indicar que un verbo está en infinitivo)	
toaster	s.	tostadora	
today	adv	hoy	
tomorrow	adv	mañana	s mañana
ton	s	tonelada	
tool	s	herramienta	

tooth	s	diente		
top	s	parte superior		
topic	s.	tema		
toward	prep	hacia		
transfer	s	transferencia	v	transferir
transform	v	transformar		
transportation	s	transporte		
try	v	probar, intentar		
tube	s	tubo		
turbine	s	turbina		
TV set	s	televisor		
two	num	dos		
type	v	escribir (a máquina), tipear	s	tipo
typical	adj	típico		

U

ultimate	adj	final		
under	prep	bajo, debajo de		
understand	v	entender		
undesirable	adj.	indeseable		
uniform	s	uniforme		
unimportant	adj	no importante, sin importancia		
university	s	universidad		
until	conj	hasta	prep	hasta
upper	adj	superior		
uranium	s	uranio		
us	pron obj	nos, nosotros, nosotras		
usage	s.	uso		
use	s	uso		
useful	adj	útil		
user	s	usuario		
user input	s	contribuciones de los usuarios, datos de entrada de usuario		
utilization	s	utilización		

V

vacuum	s.	vacío		
value	s	valor		
variable	s	variable	adj	variable
various	adj.	distinto, diferente		
vary	v.	variar		
vast	adj	vasto		
vertical	adj	vertical		
very	adv	muy		
vessel	s	recipiente, vasija, navío		
visualize	v	visualizar		
vocabulary	s	vocabulario		
voltage	s	tensión (voltaje)		

W

want	v	querer		
was	v	pretérito del verbo to be (pasado de am, is)		
variable	s	variable	adj	variable
warm	s.	calentar		
water	s	agua		
waterwheel	s	rueda hidráulica		
way	s	modo, manera, vía, camino		

web	s	red		
week	s	semana		
weight	s	peso, pesa		
well	s	pozo	adv	bien
were	v	pretérito del verbo to be (pasado de are)		
what	pron rel	lo que,	pron interr.	qué, cual
when	adv	cuando	pron interr.	cuando
where	adv	donde, adonde		
which	pron	que, el cual/la cual	pron interr	cuál
why	pron	por qué		
window	s	ventana		
wire	s	cable, alambre		
wish	v	desear	s	deseo
with	prep	con		
without	prep	sin		
woman	s	mujer		
wood	s	madera		
word	s	palabra		
work	v	trabajar, funcionar	s	trabajo
world	s	mundo		
worldwide	adj	mundial		
worth	adj	valioso, valuado		
write	v	escribir (irreg. wrote, written)		

Y

year	s	año
yesterday	adv	ayer
your	adj pos	su (de Ud., de Uds.)
yourself	pron	vos/Ud. mismo

Z

zero	num	cero
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LISTA DE VERBOS IRREGULARES

PRESENTE	PRETÉRITO	PARTICPIO PASADO	SIGNIFICADO
become	became	become	hacerse, volverse, convertirse
begin	began	begun	empezar
break	broke	broken	romper
bring	brought	brought	traer
burn	burned, burnt	burned, burnt	quemar-se
buy	bought	bought	comprar
choose	chose	chosen	elegir
come	came	come	venir
do	did	done	hacer
draw	drew	drawn	dibujar, extraer, trasfilar
fall	fell	fallen	caer-se
find	found	found	encontrar
forget	forgot	forgotten, forgot	olvidar-se
get	got	got, gotten	obtener, conseguir, etc.
give	gave	given	dar
go	went	gone	ir
grow	grew	grown	crecer
have	had	had	tener
hold	held	held	mantener, fijar
keep	kept	kept	mantener, conservar
know	knew	known	saber, conocer
learn	learned, learnt	learned, learnt	aprender
leave	left	left	salir, dejar
make	made	made	hacer
meet	met	met	reunir-se, encontrar-se
put	put	put	poner, colocar
run	ran	run	correr, ejecutar, hacer funcionar
see	saw	seen	ver
send	sent	sent	enviar
show	showed	shown, showed	mostrar
slide	slid	slid, slidden	deslizar-se
take	took	taken	tomar, llevar
teach	taught	taught	enseñar
tell	told	told	decir (a)
think	thought	thought	pensar
understand	understood	understood	entender
write	wrote	written	escribir; "write down": anotar

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